

Chapter 3

Cements

The purposes of this chapter are to present (1) the primary objectives of cementing, (2) the test procedures used to determine if the cement slurry and set cement have suitable properties for meeting these objectives, (3) the common additives used to obtain the desirable properties under various well conditions, and (4) the techniques used to place the cement at the desired location in the well. The mathematical modeling of the flow behavior of the cement slurry is not discussed in this chapter but is presented in detail in Chap. 4.

Cement is used in the drilling operation to (1) protect and support the casing, (2) prevent the movement of fluid through the annular space outside the casing, (3) stop the movement of fluid into vugular or fractured formations, and (4) close an abandoned portion of the well. A cement slurry is placed in the well by mixing powdered cement and water at the surface and pumping it by hydraulic displacement to the desired location. Thus, the hardened, or reacted, cement slurry becomes “set” cement, a rigid solid that exhibits favorable strength characteristics.

The drilling engineer is concerned with the selection of the best cement composition and placement technique for each required application. A deep well that encounters abnormally high formation pressure may require several casing strings to be cemented properly in place before the well can be drilled and completed successfully. The cement composition and placement technique for each job must be chosen so that the cement will achieve an adequate strength soon after being placed in the desired location. This minimizes the waiting period after cementing. However, the cement must remain pumpable long enough to allow placement to the desired location. Also, each cement job must be designed so that the density and length of the unset cement column results in sufficient subsurface

pressure to control the movement of pore fluid while not causing formation fracture. Consideration must be given to the composition of subsurface contaminating fluids to which the cement will be exposed.

The main ingredient in almost all drilling cements is *portland cement*, an artificial cement made by burning a blend of limestone and clay. This is the same basic type of cement used in making concrete. A slurry of portland cement in water is ideal for use in wells because it can be pumped easily and hardens readily in an underwater environment. The name “portland cement” was chosen by its inventor, Joseph Aspdin, because he thought the produced solid resembled a stone quarried on the Isle of Portland off the coast of England.

3.1 Composition of Portland Cement

A schematic representation of the manufacturing process for portland cement is shown in Fig. 3.1. The oxides of Ca, Al, Fe, and Si react in the extreme temperature of the kiln (2600 to 2800°F), resulting in balls of cement *clinker* upon cooling. After aging in storage, the seasoned clinker is taken to the grinding mills where gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) is added to retard setting time and increase ultimate strength. The unit sold by the cement company is the *barrel*, which contains 376 lbm or four 94-lbm sacks.

Cement chemists feel that there are four crystalline compounds in the clinker that hydrate to form or aid in the formation of a rigid structure. These are (1) tricalcium silicate ($3\text{CaO} \cdot \text{SiO}_2$ or “ C_3S ”), (2) dicalcium silicate ($2\text{CaO} \cdot \text{SiO}_2$ or “ C_2S ”), (3) tricalcium aluminate ($3\text{CaO} \cdot \text{Al}_2\text{O}_3$ or “ C_3A ”), and (4) tetracalcium aluminoferrite ($4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3$ or “ C_4AF ”). The hydration reaction is exothermic and generates a considerable quantity of heat, especially the hydration of C_3A .

The chemical equations representing the hydration

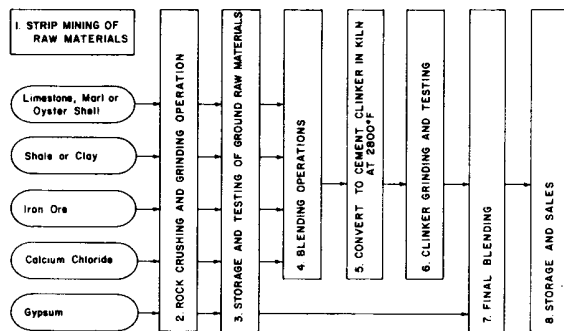
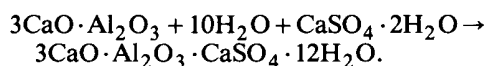
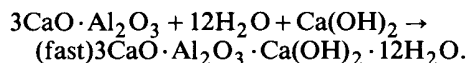
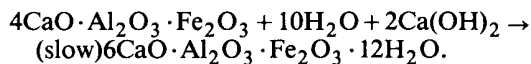
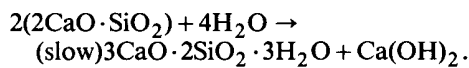
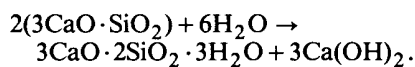


Fig. 3.1 – Manufacture of Portland cement.

of the cement compounds when they are mixed with water are as follows.



The main cementing compound in the reaction products is $3\text{CaO} \cdot 2\text{SiO}_2 \cdot 3\text{H}_2\text{O}$, which is called tobermorite gel. The gel has an extremely fine particle size and, thus, a large surface area. Strong surface attractive forces causes the gel to adsorb on all crystals and particles and bind them together. Excess water that is not hydrated reduces cement strength and makes the cement more porous and permeable.

C_3S is thought to be the major contributor to strength, especially during the first 28 days of curing. C_2S hydrates very slowly and contributes mainly to the long term strength. C_3A hydrates very rapidly and produces most of the heat of hydration observed during the first few days. The gypsum added to the clinker before grinding controls the rapid hydration of C_3A . The C_3A portion of the cement also is attacked readily by water containing sulfates. C_4AF has only minor effects on the physical properties of the cement.

The chemical composition of portland cement generally is given in terms of oxide analysis. The relative amounts of the four crystalline compounds present are computed from the oxide analysis. API¹ uses the following equations for calculating the weight percent of the crystalline compounds from the weight percent of the oxides present.

$$\text{C}_3\text{S} = 4.07\text{C} - 7.6\text{S} - 6.72\text{A} - 1.43\text{F} - 2.85\text{SO}_3. \quad (3.1)$$

$$\text{C}_2\text{S} = 2.87\text{S} - 0.754\text{C}_3\text{S}. \quad (3.2)$$

$$\text{C}_3\text{A} = 2.65\text{A} - 1.69\text{F}. \quad (3.3)$$

$$\text{C}_4\text{AF} = 3.04\text{F}. \quad (3.4)$$

These equations are valid as long as the weight ratio of Al_2O_3 to Fe_2O_3 present is greater than 0.64.

Example 3.1. Calculate the percentages of C_3S , C_2S , C_3A , and C_4AF from the following oxide analysis of a standard portland cement.

Oxide	Weight Percent
Lime (CaO or C)	65.6
Silica (SiO_2 or S)	22.2
Alumina (Al_2O_3 or A)	5.8
Ferric oxide (Fe_2O_3 or F)	2.8
Magnesia (MgO)	1.9
Sulfur trioxide (SO_3)	1.8
Ignition loss	0.7

Solution. The A/F ratio is $5.8/2.8 = 2.07$. Thus, using Eqs. 3.1 through 3.4 yields

$$\begin{aligned} \text{C}_3\text{S} &= 4.07(65.6) - 7.6(22.2) - 6.72(5.8) \\ &\quad - 1.43(2.8) - 2.85(1.8) \\ &= 50.16\% . \end{aligned}$$

$$\begin{aligned} \text{C}_2\text{S} &= 2.87(22.2) - 0.754(50.16) \\ &= 25.89\% . \end{aligned}$$

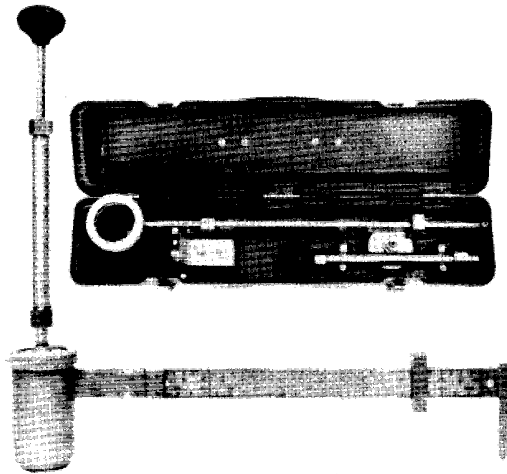
$$\begin{aligned} \text{C}_3\text{A} &= 2.65(5.8) - 1.69(2.8) \\ &= 10.64\% . \end{aligned}$$

$$\begin{aligned} \text{C}_4\text{AF} &= 3.04(2.8) \\ &= 8.51\% . \end{aligned}$$

3.2 Cement Testing

API¹ presents a recommended procedure for testing drilling cements. These tests were devised to help drilling personnel determine if a given cement composition will be suitable for the given well conditions. Cement specifications almost always are stated in terms of these standard tests. The test equipment needed to perform the API tests includes: (1) a *mud balance* for determining the slurry density, (2) a *filter press* for determining the filtration rate of the slurry, (3) a *rotational viscometer* for determining the rheological properties of the slurry, (4) a *consistometer* for determining the thickening rate characteristics of the slurry, (5) a *cement permeameter* for determining the permeability of the set cement, (6) *specimen molds* and *strength testing machines* for determining the tensile and compressive strength of the cement, (7) an *autoclave* for determining the soundness of the cement, and (8) a *turbidimeter* for determining the fineness of the cement. Unlike drilling fluid testing, routine testing of the cement slurry normally is not done at the rig site. However, it is imperative for the drilling engineer to understand the nature of these tests if he is to interpret cement specifications and reported test results properly.

The mud balance, filter press, and rotational viscometer used for cement testing are basically the same equipment described in Chap. 2 for testing drilling fluids. However, when measuring the density of cement slurries, entrained air in the sample is more difficult to remove. The pressurized mud balance



Courtesy of Halliburton Services

Fig. 3.2 - Pressurized mud balance.

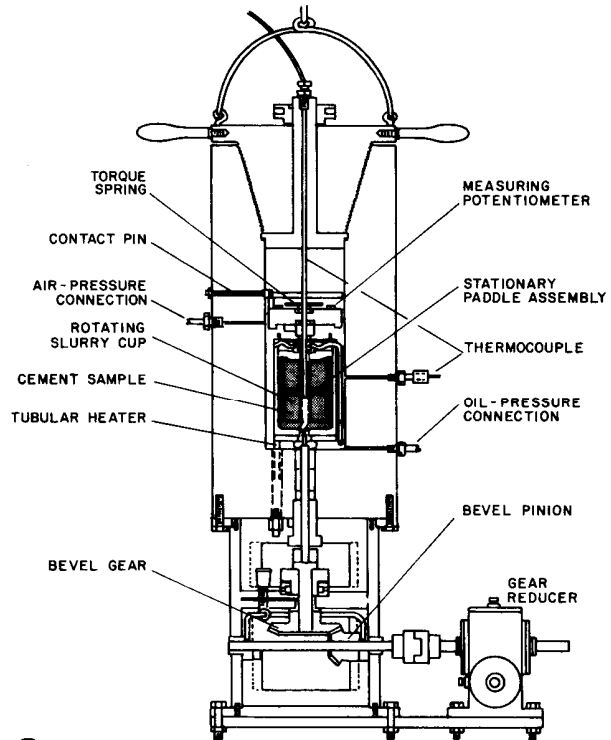
shown in Fig. 3.2 can be used to minimize the effect of the entrained air.

3.2.1 Cement Consistometer. The pressurized and atmospheric-pressure consistometers used in testing cement are shown in Fig. 3.3. The apparatus consists essentially of a rotating cylindrical slurry container equipped with a stationary paddle assembly, all enclosed in a pressure chamber capable of withstanding temperatures and pressures encountered in well cementing operations. The cylindrical slurry chamber is rotated at 150 rpm during the test. The slurry consistency is defined in terms of the torque exerted on the paddle by the cement slurry. The relation between torque and slurry consistency is given by

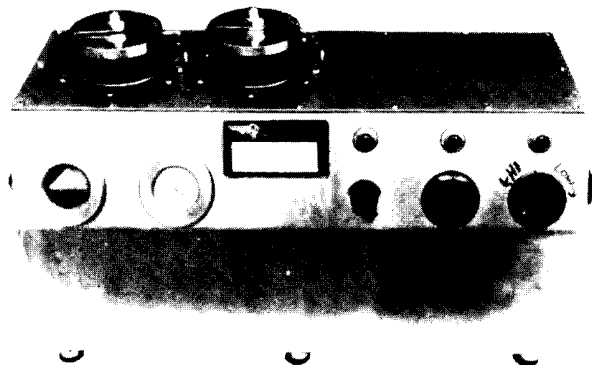
$$B_c = \frac{T - 78.2}{20.02}, \dots \dots \dots (3.5)$$

where T = the torque on the paddle in g-cm and B_c = the slurry consistency in API consistency units designated by B_c . The thickening time of the slurry is defined as the time required to reach a consistency of 100 B_c . This value is felt to be representative of the upper limit of pumpability. The temperature and pressure schedule followed during the test must be given with the thickening time for the test results to be meaningful. API periodically reviews field data concerning the temperatures and pressures encountered during various types of cementing operations and publishes recommended schedules for use with the consistometer. At present, 31 published schedules are available for simulating various cementing operations. Schedule 6, designed to simulate the average conditions encountered during the cementing of casing at a depth of 10,000 ft, is shown in Table 3.1.

The atmospheric-pressure consistometer is frequently used to simulate a given history of slurry pumping before making certain tests on the slurry. For example, the rheological properties of cement slurries are time dependent since the cement thickens with time. The history of shear rate, temperature, and pressure before measuring



a



b

Fig. 3.3 - Cement consistometer: (a) schematic of high-pressure consistometer, (b) atmospheric-pressure consistometer.

the cement rheological properties using a rotational viscometer can be specified in terms of a schedule followed using the consistometer. The consistometer also is used to determine the maximum, minimum, normal, and free water content of the slurry. In these tests, the sample is placed first in the consistometer and stirred for a period of 20 minutes at 80°F and atmospheric pressure. The *minimum water content* is the amount of mixing water per sack of cement that will result in a consistency of 30 B_c at the end of this period. The *normal water content* is the amount of mixing water per sack of cement that will result in a consistency of 11 B_c at the end of the test. The *free water content* is determined by pouring a 250-mL sample from the consistometer into a

TABLE 3.1 – EXAMPLE CONSISTOMETER SCHEDULE¹
(Schedule 6 – 10,000-ft (3050 m) casing cement
specification test)

Surface temperature, °F (°C)	80 (27)
Surface pressure, psi (kg/cm ²)	1,250 (88)
Mud density	
lbm/gal (kg/L)	12 (1.4)
lbm/cu ft	89.8
psi/Mft (kg/cm ³ /m)	623 (0.144)
Bottomhole temperature, °F (°C)	144 (62)
Bottomhole pressure, psi (kg/cm ²)	7,480 (526)
Time to reach bottom, minutes	36

Time (minutes)	Pressure		Temperature	
	(psi)	(kg/cm ²)	(°F)	(°C)
0	1,250	88	80	27
2	1,600	113	84	29
4	1,900	134	87	31
6	2,300	162	91	33
8	2,600	183	94	34
10	3,000	211	98	37
12	3,300	232	101	38
14	3,700	260	105	41
16	4,000	281	108	42
18	4,400	309	112	44
20	4,700	330	116	47
22	5,100	359	119	48
24	5,400	380	123	51
26	5,700	401	126	52
28	6,100	429	130	54
30	6,400	451	133	56
32	6,800	478	137	58
34	7,100	499	140	60
36	7,480	526	144	62

Final temperature and pressure should be held constant to completion of test, within $\pm 2^\circ\text{F}$ ($\pm 1^\circ\text{C}$) and ± 100 psi (± 7 kg/cm²), respectively.

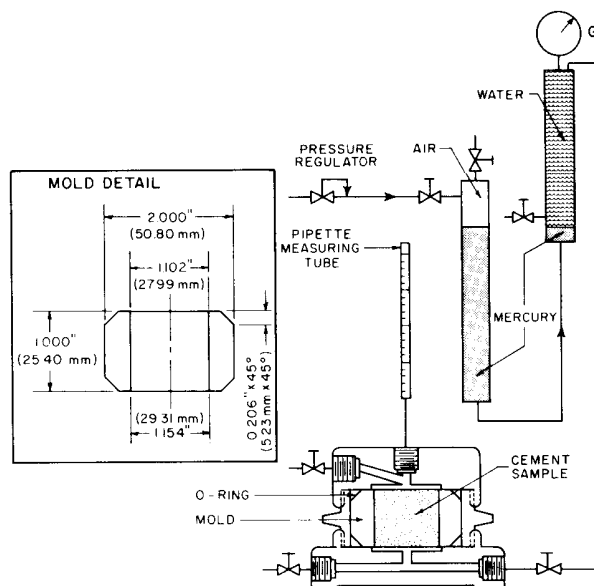


Fig. 3.4 – Cement permeameter.

glass graduated cylinder and noting the amount of free supernatant water that separates from the slurry over a 2-hour period. The *maximum water content* is defined as the amount of water per sack of cement that will result in 2.5 mL of free water. A consistometer designed to operate only at atmospheric pressure is frequently used in conjunction with the determination of the slurry rheological properties and water content.

Example 3.2. The torque required to hold the paddle assembly stationary in a cement consistometer rotating at 150 rpm is 520 g-cm. Compute the slurry consistency.

$$B_c = \frac{T - 78.2}{20.02} = \frac{520 - 78.2}{20.02}$$

$$= 22 \text{ consistency units.}$$

3.2.2 Cement Permeameter. A schematic of the permeameter used in the cement permeability test recommended by API is shown in Fig. 3.4.¹ The permeability of a set cement core to water is determined by measuring the flow rate through the core at

a given pressure differential across the length of the core. The permeability then is computed using an appropriate form of Darcy's law:

$$k = 14,700 \frac{q\mu L}{A\Delta p}, \dots \dots \dots (3.6)$$

where

k = permeability, md,
 q = flow rate, mL/s,
 μ = water viscosity, cp,
 L = sample length, cm,
 A = sample cross-sectional area, cm², and
 Δp = pressure differential, psi.

The curing time, temperature, and pressure of the sample usually are reported with the cement permeability.

Example 3.3. A Class E cement core having a length of 2.54 cm and a diameter of 2.865 cm allows a water flow rate of 0.0345 mL/s when placed under a pressure differential of 20 psi. A second core containing 40% silica cured in a similar manner allows only 0.00345 mL/s of water to flow under a pressure differential of 200 psi. Compute the permeability of the two cement samples.

Solution. Using Darcy's law for linear flow of liquids as defined by Eq. 3.6 gives

TABLE 3.2 – WELL SIMULATION TEST SCHEDULES FOR CURING STRENGTH SPECIMENS

Schedule Number	Depth		Pressure*		Temperature, °F (°C)											
					Elapsed Time from First Application of Heat and Pressure											
	(ft)	(m)	(psi)	(kg/cm ²)	0:30	0:45	1:00	1:15	1:30	2:00	2:30	3:00	3:30	4:00**		
1S	1,000	310	800	56	82 (28)	83 (28)	84 (29)	85 (29)	86 (30)	87 (31)	89 (31)	91 (33)	93 (34)	95 (35)		
2S	2,000	610	1,600	113	93 (34)	94 (34)	96 (36)	97 (36)	98 (37)	100 (38)	103 (39)	105 (41)	108 (42)	110 (43)		
3S	4,000	1220	3,000	211	106 (41)	108 (42)	110 (43)	113 (45)	115 (46)	120 (49)	125 (52)	130 (54)	135 (57)	140 (60)		
4S	6,000	1830	3,000	211	116 (47)	120 (49)	124 (51)	128 (53)	131 (55)	139 (59)	147 (64)	155 (68)	162 (72)	170 (77)		
5S	8,000	2440	3,000	211	126 (52)	131 (55)	136 (58)	142 (61)	147 (64)	158 (70)	168 (76)	179 (82)	190 (88)	200 (93)		
6S	10,000	3050	3,000	211	133 (56)	148 (64)	154 (68)	161 (72)	167 (75)	180 (82)	192 (89)	205 (96)	218 (103)	230 (110)		
7S	12,000	3660	3,000	211	143 (62)	173 (78)	179 (82)	186 (86)	193 (89)	206 (97)	220 (104)	233 (112)	246 (119)	260 (127)		
8S	14,000	4270	3,000	211	153 (67)	189 (87)	210 (99)	216 (103)	223 (106)	236 (113)	250 (121)	263 (128)	277 (136)	290 (143)		
9S	16,000	4880	3,000	211	164 (73)	206 (97)	248 (120)	254 (123)	260 (127)	272 (133)	284 (140)	296 (147)	308 (153)	320 (160)		
10S	18,000	5490	3,000	211	179 (82)	227 (108)	277 (136)	302 (150)	307 (153)	315 (157)	324 (162)	333 (167)	341 (172)	350 (177)		
11S	20,000	6100	3,000	211	184 (84)	236 (113)	288 (142)	340 (171)	344 (173)	351 (177)	358 (181)	366 (186)	373 (189)	380 (193)		

*The test pressure shall be applied as soon as specimens are placed in the pressure vessel and maintained at the given pressure within the following limits for the duration of the curing period:

• Schedule 1S 800 ± 100 psi (56 ± 7 kg/cm²)
 • Schedule 2S 1600 ± 200 psi (113 ± 14 kg/cm²)
 • Schedules 3S through 11S 3000 ± 500 psi (211 ± 35 kg/cm²)

**Final temperature (Col. 13) shall be maintained ± 3°F (± 2°C) throughout the remainder of the curing period.

$$k_1 = 14,700 \frac{0.0345(1.0)(2.54)}{\frac{\pi}{4}(2.865)^2 (20)} = 10 \text{ md.}$$

The permeability observed after the addition of silica is given by

$$k_2 = 14,700 \frac{0.00345(1.0)(2.54)}{\frac{\pi}{4}(2.865)^2 (200)} = 0.1 \text{ md.}$$

3.2.3 Strength, Soundness and Fineness Tests. The standard tests for cement compressive strength, tensile strength, soundness, and fineness published in the latest ASTM C 190, C 109, C 151, and C 115 also are used by API for drilling cements. The compressive strength of the set cement is the compressional force required to crush the cement divided by the cross-sectional area of the sample. Test schedules for curing strength test specimens are recommended by API. These schedules are based on average conditions encountered in different types of cementing operations and are updated periodically on the basis of current field data. The test schedules published in RP 10B¹ in Jan. 1982 are given in Table 3.2. *The compressive strength of the cement is usually about 12 times greater than the tensile strength* at any given curing time. Thus, frequently only the compressive strength is reported.

The *soundness* of the cement is the percent linear expansion or contraction observed after curing in an autoclave under saturated steam at a pressure of 295 psig for 3 hours. A cement that changes dimensions upon curing may tend to bond poorly to casing or to form cracks.

The *fineness* of the cement is a measure of the size of the cement particle achieved during grinding. The fineness is expressed in terms of a calculated total particle surface area per gram of cement. The fineness is calculated from the rate of settlement of cement particles suspended in kerosene in a Wagner turbidimeter. The finer the cement is ground during manufacture, the greater the surface area available

for contact with water, and the more rapid is the hydration process.

3.3 Standardization of Drilling Cements

API has defined eight standard classes and three standard types of cement for use in wells. The eight classes specified are designated Class A to Class H. The intended meanings of the various classes are defined in Table 3.3. The three types specified are (1) ordinary "O," (2) moderate sulfate-resistant "MSR," and (3) high sulfate-resistant "HSR." The chemical requirement for the various types and classes are given in Table 3.4 and the physical requirements are given in Table 3.5.

The physical requirements of the various classes of cement given in Table 3.5 apply to cement samples prepared according to API specifications. To provide uniformity in testing, it is necessary to specify the amount of water to be mixed with each type of cement. These water-content ratios, shown in Table 3.6, often are referred to as the normal water content or "API water" of the cement class. As will be discussed in the next section, Wyoming bentonite sometimes is added to the cement slurry to reduce the slurry density, or barium sulfate is added to increase the slurry density. API specifies that the water content be increased 5.3 wt% for each weight percent of bentonite added and 0.2 wt% for each weight percent of barium sulfate added.

The relation between well depth and cementing time used in the specifications for the various cement classes is shown in Fig. 3.5. The relation shown assumes a cement mixing time of 20 cu ft/min and a displacement rate after mixing of 50 cu ft/min. Also, a 7.0-in.-OD casing having a cross-sectional area of 33.57 sq in. is assumed. For these conditions, which are felt to be representative of current field practice, the time required to mix and displace various volumes of cement has been plotted as a function of depth. Also plotted are the cement volumes used in determining the recommended minimum thickening time.

3.3.1 Construction Industry Cement Designations.

The majority of the cement produced in this country is used in construction with only about 5% being

TABLE 3.3—STANDARD CEMENT CLASSES DESIGNATED BY API¹

Class A:	Intended for use from surface to 6,000-ft (1830-m) depth, when special properties are not required. Available only in ordinary type (similar to ASTM C 150 Type I).
Class B:	Intended for use from surface to 6,000-ft (1830-m) depth, when conditions require moderate to high sulfate-resistance. Available in both moderate (similar to ASTM C 150, Type II) and high sulfate-resistant types.
Class C:	Intended for use from surface to 6,000-ft (1830-m) depth, when conditions require high early strength. Available in ordinary and moderate (similar to ASTM C 150, Type III) and high sulfate-resistant types.
Class D:	Intended for use from 6,000- to 10,000-ft depth (1830- to 3050-m) depth, under conditions of moderately high temperatures and pressures. Available in both moderate and high sulfate-resistant types.
Class E:	Intended for use from 10,000- to 14,000-ft (3050- to 4270-m) depth, under conditions of high temperatures and pressures. Available in both moderate and high sulfate-resistant types.
Class F:	Intended for use from 10,000- to 16,000-ft (3050- to 4880-m) depth, under conditions of extremely high temperatures and pressures. Available in both moderate and high sulfate-resistant types.
Class G:	Intended for use as a basic cement from surface to 8,000-ft (2400-m) depth as manufactured, or can be used with accelerators and retarders to cover a wide range of well depths and temperatures. No additions other than calcium sulfate or water, or both, shall be interground or blended with the clinker during manufacture of Class G cement. Available in moderate and high sulfate-resistant types.
Class H:	Intended for use as a basic cement from surface to 8,000-ft (2440-m) depth as manufactured, and can be used with accelerators and retarders to cover a wide range of well depths and temperatures. No additions other than calcium sulfate or water, or both, shall be interground or blended with the clinker during manufacture of Class H cement. Available only in moderate sulfate-resistant type.

used in oil and gas wells. In some cases, it may be necessary to use cement products normally marketed for the construction industry. This is especially true when working in foreign countries. Five basic types of portland cements are used commonly in the construction industry. The ASTM classifications and international designations for these five cements are shown in Table 3.7. Note that ASTM Type I, called normal, ordinary, or common cement, is similar to API Class A cement. Likewise, ASTM Type II, which is modified for moderate sulfate resistance is similar to API Class B cement. ASTM Type III, called high early strength cement, is similar to API Class C cement.

3.4 Cement Additives

Today more than 40 chemical additives are used with various API classes of cement to provide acceptable

slurry characteristics for almost any subsurface environment. Essentially all of these additives are free-flowing powders that either can be dry blended with the cement before transporting it to the well or can be dispersed in the mixing water at the job site. At present, the cement Classes G and H can be modified easily through the use of additives to meet almost any job specifications economically. The use of a modified Class H cement has become extremely popular.

The cement additives available can be subdivided into these functional groups: (1) density control additives, (2) setting time control additives, (3) lost circulation additives, (4) filtration control additives, (5) viscosity control additives, and (6) special additives for unusual problems. The first two categories are perhaps the most important because they receive consideration on almost every cement job. Some additives serve more than one purpose and, thus, would fit under more than one of the classifications shown above.

The nomenclature used by the petroleum industry to express the concentration of cement additives often is confusing to the student. However, most of the confusion can be cleared up by pointing out that the reference basis of cement mixtures is a unit weight of cement. When the concentration of an additive is expressed as a "weight percent" or just "percent," the intended meaning is usually that the weight of the additive put in the cement mixture is computed by multiplying the weight of cement in the mixture by the weight percent given by 100%. The concentration of liquid additives sometimes is expressed as gallons per sack of cement. A sack of cement contains 94 lbm unless the cement product is a blend of cement and some other material. The water content of the slurry sometimes is expressed as water cement ratio in gallons per sack and sometimes expressed as a weight percent. The term "percent mix" is used for water content expressed as a weight percent. Thus,

$$\text{percent mix} = \frac{\text{water weight}}{\text{cement weight}} \times 100.$$

The theoretical volume of the slurry mixture is calculated using the same procedure outlined in Sec. 2.2 of Chap. 2 for drilling fluids. Ideal mixing can be assumed unless one or more of the components are dissolved in the water phase of the cement. Many components are used in low concentration and have very minor effects on slurry volume. Physical properties of cement components needed to perform the ideal mixing calculations are given in Table 3.8.

The volume of slurry obtained per sack of cement used is called the *yield* of the cement. This term should not be confused with the yield of a clay or the yield point of a fluid as discussed in Chap. 2.

Example 3.4. It is desired to mix a slurry of Class A cement containing 3% bentonite, using the normal mixing water as specified by API (Table 3.6). Determine the weight of bentonite and volume of

TABLE 3.4—CHEMICAL REQUIREMENTS OF API CEMENT TYPES¹

Ordinary Type (O)	Cement Class					
	A	B	C	D,E,F	G	H
Magnesium oxide (MgO), maximum, %	5.00	—	5.00	—	—	—
Sulfur trioxide (SO ₃), maximum, %	3.50	—	4.50	—	—	—
Loss on ignition, maximum, %	3.00	—	3.00	—	—	—
Insoluble residue, maximum, %	0.75	—	0.75	—	—	—
Tricalcium aluminate (3CaO·Al ₂ O ₃), maximum, %	—	—	15.00	—	—	—
Moderate Sulfate-Resistant Type (MSR)						
Magnesium oxide (MgO), maximum, %	—	5.00	5.00	5.00	5.00	5.00
Sulfur trioxide (SO ₃), maximum, %	—	3.00	3.50	2.50	2.50	2.50
Loss on ignition, maximum, %	—	3.00	3.00	3.00	3.00	3.00
Insoluble residue, maximum, %	—	0.75	0.75	0.75	0.75	0.75
Tricalcium silicate (3CaO·SiO ₂), %						
maximum	—	—	—	—	58.00	58.00
maximum	—	—	—	—	48.00	48.00
Tricalcium aluminate (3CaO·Al ₂ O ₃), maximum, %	—	8.00	8.00	8.00	8.00	8.00
Total alkali content expressed as sodium oxide (Na ₂ O) equivalent, maximum, %	—	—	—	—	0.60	0.60
High Sulfate-Resistant Type (HSR)						
Magnesium oxide (MgO), maximum, %	—	5.00	5.00	5.00	5.00	—
Sulfur trioxide (SO ₃), maximum, %	—	3.00	3.50	2.50	2.50	—
Loss on ignition, maximum, %	—	3.00	3.00	3.00	3.00	—
Insoluble residue, maximum, %	—	0.75	0.75	0.75	0.75	—
Tricalcium silicate (3CaO·SiO ₂), %						
maximum	—	—	—	—	65.00	—
maximum	—	—	—	—	48.00	—
Tricalcium aluminate (3CaO·Al ₂ O ₃), maximum, %	—	3.00	3.00	3.00	3.00	—
Tetracalcium aluminoferrite (4CaO·Al ₂ O ₃ ·Fe ₂ O ₃) plus twice the tricalcium aluminate (3CaO·Al ₂ O ₃), maximum, %	—	24.00	24.00	24.00	24.00	—
Total alkali content expressed as sodium oxide (Na ₂ O) equivalent, maximum, % ³					0.60	

TABLE 3.5—PHYSICAL REQUIREMENTS OF API CEMENT TYPES¹

	Cement Case							
	A	B	C	D	E	F	G	H
Soundness (autoclave expansion) maximum, %	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80
Fineness* (specific surface), minimum, cm ² /g	1,500	1,600	2,200	—	—	—	—	—
Free water content, maximum, mL	—	—	—	—	—	—	3.5**	2.5**
Compressive Strength								
Test	Schedule Number, Table 6.1 RP10B	Curing Temperature		Curing Pressure		Minimum Compressive Strength, psi (kg/cm ²)		
		(°F)	(°C)	(psi)	(kg/cm ²)			
8-Hour	—	100	38	Atmos.	250 (18)	200 (14)	300 (21)	—
Curing	1S	95	35	800	56	—	—	—
Time	3S	140	60	3,000	211	—	—	—
	6S	230	110	3,000	211	—	500 (35)	—
	8S	290	143	2,000	211	—	500 (35)	—
	9S	330	2,000	211	—	—	500 (35)	—
Compressive Strength								
Test	Schedule Number, Table 6.1 RP10B	Curing Temperature		Curing Pressure		Minimum Compressive Strength, psi (kg/cm ²)		
		(°F)	(°C)	(psi)	(kg/cm ²)			
24-Hour	—	100	38	Atmos.	1,800 (127)	1,500 (106)	2,000 (141)	—
Curing	4S	170	77	2,000	211	—	1,000 (70)	1,000 (70)
Time	6S	230	110	3,000	211	—	2,000 (141)	1,000 (70)
	8S	290	143	3,000	211	—	2,000 (141)	—
	9S	320	160	2,000	211	—	1,000 (70)	—
Pressure								
Temp-erature	Well Simulation Test	Maximum Consistency		Minimum Thickening Time (minutes)***				
	Schedule Number, Table 7.2 RP10B	Simulated Well Depth (ft)	(m)					
Thick-ening	1	1,000	310	30	90	90	90	—
Time	4	6,000	1830	30	90	90	90	—
Test	5	8,000	2440	30	—	—	—	90
	6	10,000	3050	30	—	—	100	100
	8	14,000	4270	30	—	—	154	—
	9	16,000	4880	30	—	—	190	—

*Determined by Wagner turbidimeter apparatus described in ASTM C 115 *Fineness of Portland Cement by the Turbidimeter*, current edition of *ASTM Book of Standards*, Part 9.

**Based on 250 mL volume, percentage equivalent of 3.5 mL is 1.4%.

***Thickening time requirements are based on 75 percentile values of the total cementing times observed in the casing survey, plus a 25% safety factor.

†Units of slurry consistency (U) formerly referred to as "poises."

‡Maximum thickening-time requirement for Schedule 5 is 120 minutes.

TABLE 3.6—NORMAL WATER CONTENT OF CEMENT RECOMMENDED BY API

API Class Cement	Water (%) by Weight of Cement	Water	
		gal per sack	L per sack
A and B	46	5.19	19.6
C	56	6.32	23.9
D, E, F, and H	38	4.29	16.2
G	44	4.97	18.8
J (tentative)	*	*	*

*As recommended by the manufacturer.

Note 1: The addition of bentonite to cement requires that the amount of water be increased. It is recommended, for testing purposes, that 5.3% water be added for each 1% bentonite in all API classes of cement. For example, a Class A cement slurry having a water/cement ratio of 0.46, to which is added 3% bentonite, will require an increase in water/cement ratio to 0.619.

Note 2: The addition of barite to cement generally requires that the amount of water be increased. It is recommended, for testing purposes, that 0.2% water be added for each 1% barite. For example, a cement slurry having a normal water/cement ratio of 0.38 and weighted to 18 lbm/gal (134.6 lbm/cu ft) (2.2 kg/L) by addition of 60% barite, will require an increase in water/cement ratio to 0.50.

water to be mixed with one 94-lbm sack of cement. Also, compute the percent mix, yield, and density of the slurry.

Solution. The weight of bentonite to be blended with one sack of Class A cement is

$$0.03(94) = 2.82 \text{ lbm.}$$

The normal water content for Class A cement is 46% (Table 3.6). However, 5.3% water must be added for each percent bentonite. Thus, the percent mix is

$$46 + 3(5.3) = 61.9\%.$$

The water volume to be added per sack of Class A cement is given by

$$\frac{0.619(94 \text{ lbm/sack})}{8.33 \text{ lbm/gal}} = 6.98 \text{ gal.}$$

The specific gravities of cement and bentonite are 3.14 and 2.65, respectively (Table 3.8). The volume of the slurry is given by

$$\frac{94 \text{ lbm}}{3.14(8.33) \text{ lbm/gal}} + \frac{2.82 \text{ lbm}}{2.65(8.33) \text{ lbm/gal}} + 6.98 \text{ gal/sack} = 10.7 \text{ gal/sack.}$$

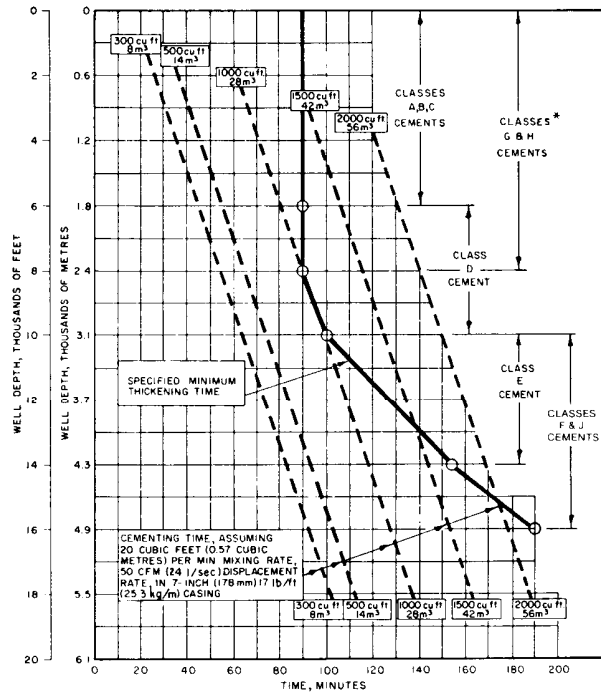


Fig. 3.5 – Well depth and cementing time relationship used in definition of API cement classes.¹

The yield of the slurry is

$$\frac{10.7 \text{ gal/sack}}{7.48 \text{ gal/cu ft}} = 1.43 \text{ cu ft/sack.}$$

The density of the slurry is the total mass divided by the total volume or

$$\frac{94 + 2.82 + 8.33(6.98)}{10.7} = 14.48 \text{ lbm/gal.}$$

3.4.1 Density Control. The density of the cement slurry must be high enough to prevent the higher-pressured formations from flowing into the well during cementing operations, yet not so high as to cause fracture of the weaker formations. In most cases, the density of the cement slurry obtained by mixing cement with the normal amount of water will be too great for the formation fracture strength, and it will be desirable to lower the slurry density.

TABLE 3.7—BASIC ASTM CEMENT TYPES²

ASTM Type	International Designation	API Class	Common Name	Typical Composition			
				C ₃ S	C ₂ S	C ₃ A	C ₄ AF
I	OC	A	normal, ordinary, or common	53	24	8	8
II		B	modified	47	32	3	12
III	RHC	C	high early strength	58	16	8	8
IV	LHC		low heat	26	54	2	12
V	SRC		sulfate- resisting	—	—	—	—

Reducing the cement density also tends to reduce the overall cost of the cement slurry. Slurry density is reduced by using a higher water/cement ratio or adding low-specific-gravity solids, or both. Also, nonstandard cements having a lower specific gravity are available. Trinity Lite-wate™ cement, which has a specific gravity of 2.8, is a very popular low-density cement.

The low-specific-gravity solids commonly used to reduce slurry density include (1) bentonite (sodium montmorillonite), (2) diatomaceous earth, (3) solid hydrocarbons, (4) expanded perlite, and (5) poz-zolan. When extremely weak formations are present, it may not be possible to reduce slurry density sufficiently to prevent fracture. In this case, the mud column in front of the cement slurry can be aerated

TABLE 3.8—PHYSICAL PROPERTIES OF CEMENTING MATERIALS⁵

Material	Bulk Weight (lbm/cu ft)	Specific Gravity	Weight 3.6 ^a Absolute Gal	Absolute Volume	
				gal/lbm	cu ft/lbm
API cements	94	3.14	94	0.0382	0.0051
Ciment Fondu	90	3.23	97	0.0371	0.0050
Lumnite cement	90	3.20	96	0.0375	0.0050
Trinity Lite-Wate	75	2.80	75.0 ^e	0.0429	0.0057
Activated charcoal	14	1.57	47.1	0.0765	0.0102
Barite	135	4.23	126.9	0.0284	0.0038
Bentonite (gel)	60	2.65	79.5	0.0453	0.0060
Calcium chloride, flake ^b	56.4	1.96	58.8	0.0612	0.0082
Calcium chloride, powder ^b	50.5	1.96	58.8	0.0612	0.0082
Cal-Seal, gypsum cement	75	2.70	81.0	0.0444	0.0059
CFR-1 ^b	40.3	1.63	48.9	0.0736	0.0098
CFR-2 ^b	43.0	1.30	39.0	0.0688	0.0092
DETA (liquid)	59.5	0.95	28.5	0.1258	0.0168
Diacel A ^b	60.3	2.62	78.6	0.0458	0.0061
Diacel D	16.7	2.10	63.0	0.0572	0.0076
Diacel LWL ^b	29.0	1.36	40.8	0.0882	0.0118
Diesel Oil No. 1 (liquid)	51.1	0.82	24.7	0.1457	0.0195
Diesel Oil No. 2 (liquid)	53.0	0.85	25.5	0.1411	0.0188
Gilsonite	50	1.07	32	0.1122	0.0150
HALDAD®-9 ^b	37.2	1.22	36.6	0.0984	0.0131
HALDAD®-14 ^b	39.5	1.31	39.3	0.0916	0.0122
Hematite	193	5.02	150.5	0.0239	0.0032
HR-4 ^b	35	1.56	46.8	0.0760	0.0103
HR-7 ^b	30	1.30	39	0.0923	0.0123
HR-12 ^b	23.2	1.22	36.6	0.0984	0.0131
HR-L (liquid) ^b	76.6	1.23	36.9	0.0976	0.0130
Hydrated lime	31	2.20	66	0.0545	0.0073
Hydromite	68	2.15	64.5	0.0538	0.0072
LA-2 Latex (liquid)	68.5	1.10	33	0.1087	0.0145
LAP-1 Latex ^b	50	1.25	37.5	0.0960	0.0128
LR-11 Resin (liquid)	79.1	1.27	38.1	0.0945	0.0126
NF-1 (liquid) ^b	61.1	0.98	29.4	0.1225	0.0164
NF-P ^b	40	1.30	39.0	0.0923	0.0123
Perlite regular	8 ^c	2.20	66.0	0.0546	0.0073
Perlite Six	38 ^d	—	—	0.0499	0.0067
Pozmix® A	74	2.46	74	0.0487	0.0065
Pozmix® D	47	2.50	73.6	0.0489	0.0065
Salt (dry NaCl)	71	2.17	65.1	0.0553	0.0074
Salt (in solution at 77°F with fresh water)					
6%, 0.5 lbm/gal	—	—	—	0.0384	0.0051
12%, 1.0 lbm/gal	—	—	—	0.0399	0.0053
18%, 1.5 lbm/gal	—	—	—	0.0412	0.0055
24%, 2.0 lbm/gal	—	—	—	0.0424	0.0057
Saturated, 3.1 lbm/gal	—	—	—	0.0445	0.0059
Salt (in solution at 140°F with fresh water)					
saturated, 3.1 lbm/gal	—	—	—	0.0458	0.0061
Sand (Ottawa)	100	2.63	78.9	0.0456	0.0061
Silica flour (SSA-1)	70	2.63	78.9	0.0456	0.0061
Coarse silica (SSA-2)	100	2.63	78.9	0.0456	0.0061
Tuf Additive No. 1	—	1.23	36.9	0.0976	0.0130
Tuf-Plug	48	1.28	38.4	0.0938	0.0125
Water	62.4	1.00	30.0	0.1200	0.0160

^aEquivalent to one 94-lbm sack of cement in volume.

^bWhen less than 5% is used, these chemicals may be omitted from calculations without significant error.

^cFor 8 lbm of Perlite regular use a volume of 1.43 gal at zero pressure.

^dFor 38 lbm of Perlite Six use a volume of 2.89 gal at zero pressure.

^e75 lbm = 3.22 absolute gal.

TABLE 3.9—WATER REQUIREMENTS OF CEMENTING MATERIALS⁴

Material	Water Requirements
API Class A and B cements	5.2 gal (0.70 cu ft)/94-lbm sack
API Class C cement (Hi Early)	6.3 gal (0.84 cu ft)/94-lbm sack
API Class D and E cements (retarded)	4.3 gal (0.58 cu ft)/94-lbm sack
API Class G cement	5.0 gal (0.67 cu ft)/94-lbm sack
API Class H cement	4.3 to 5.2 gal/94-lbm sack
Chem Comp cement	6.3 gal (0.84 cu ft)/94-lbm sack
Ciment Fondu	4.5 gal (0.60 cu ft)/94-lbm sack
Lumnite cement	4.5 gal (0.60 cu ft)/94-lbm sack
HLC	7.7 to 10.9 gal/87-lbm sack
Trinity Lite-Wate cement	7.7 gal (1.03 cu ft)/75-lbm sack (maximum)
Activated charcoal	none at 1 lbm/sack of cement
Barite	2.4 gal (0.32 cu ft)/100-lbm sack
Bentonite (gel)	1.3 gal (0.174 cu ft)/2% in cement
Calcium chloride	none
Gypsum hemihydrate	4.8 gal (0.64 cu ft)/100-lbm sack
CFR-1	none
CFR-2	none
Diacel A	none
Diacel D	3.3 to 7.2 gal/10% in cement (see Lt. Wt. Cement)
Diacel LWL	none (up to 0.7%) 0.8 to 1.0 gal/1% in cement (except gel or Diacel D slurries)
Gilsonite	2.0 gal (0.267 cu ft)/50 lbm/cu ft
HALAD -9	none (up to 0.5%) 0.4 to 0.5 gal/sack of cement at over 0.5%
HALAD -14	none
Hematite	0.36 gal (0.048 cu ft)/100-lbm sack
HR-4	none
HR-7	none
HR-12	none
HR-20	none
Hydrated lime	0.153 gal (0.020 cu ft)/lbm
Hydromite	3.0 gal (0.40 cu ft)/100-lbm sack
LA-2 Latex	0 to 0.8 gal/sack of cement
LAP-1 powdered latex	1.7 gal (0.227 cu ft)/1% in cement
NF-P	none
Perlite regular	4.0 gal (0.535 cu ft)/8 lbm/cu ft
Perlite Six	6.0 gal (0.80 cu ft)/38 lbm/cu ft
Pozmix A	3.6 gal (0.48 cu ft)/74 lbm/cu ft
Salt (NaCl)	none
Sand, Ottawa	none
Silica flour (SSA-1)	1.5 gal (0.20 cu ft)/35% in cement (32.9 lbm)
Coarse silica (SSA-2)	none
Tuf Additive No. 1	none
Tuf Plug	none

with nitrogen to reduce hydrostatic pressure further.

In areas where the formation pore pressure is extremely high, it may be necessary to increase the slurry density. Slurry density usually is increased by using a lower water content or adding high-specific-gravity solids. The high-specific-gravity solids commonly used to increase slurry density include (1) hematite, (2) ilmenite, (3) barite (barium sulfate), and (4) sand. The specific gravity of selected cement additives are shown in Table 3.8. The water requirements for the various additives are shown in Table 3.9.

3.4.2 Bentonite. The use of bentonite (sodium montmorillonite) clay for building drilling fluid viscosity has been discussed previously in Chap. 2. This same clay mineral is used extensively as an additive for lowering cement density. However, bentonite marketed for use in drilling fluid sometimes is treated with an organic polymer that is

undesirable for use in cement slurries since it tends to increase slurry viscosity. The addition of bentonite lowers the slurry density because of its lower specific gravity and because its ability to hydrate permits the use of much higher water concentrations. Bentonite concentrations as high as 25% by weight of cement have been used. The bentonite usually is blended dry with the cement before mixing with water, but it can be prehydrated in the mixing water. Much higher increases in water content can be obtained for each percent bentonite added when the bentonite is prehydrated in the mixing water. The ratio of bentonite dry blended to bentonite prehydrated is about 3.6:1 for comparable slurry properties.

In addition to lowering slurry density, the addition of bentonite lowers slurry cost. However, a high percentage of bentonite in cement also will cause a reduction in cement strength and thickening time. Also, the higher water content lowers the resistance to sulfate attack and increases the permeability of the set cement. At temperatures above 230°F, the use of bentonite promotes retrogression of strength in cements with time. Typical data showing the effect of bentonite concentration on various properties of Class A cement are shown in Table 3.10. However, test results have been found to vary significantly from batch to batch. When exact data are needed, tests should be conducted using the same materials and mixing water that will be used in the cementing operations.

3.4.3 Diatomaceous Earth. A special grade of diatomaceous earth also is used in portland cements to reduce slurry density. The diatomaceous earth has lower specific gravity than bentonite (Table 3.8) and permits higher water/cement ratios without resulting in free water. In addition, the silica contained in the diatomite reacts chemically with the calcium hydroxide released as portland cement sets, and it produces a gel that becomes cementitious with age and temperature. Diatomaceous earth concentrations as high as 40% by weight of cement have been used. As in the case of bentonite, the thickening time and strength of the cement are reduced as the diatomaceous earth concentration is increased. Example slurry properties obtained with various concentrations of diatomaceous earth are shown in Table 3.11.

3.4.4 Solid Hydrocarbons. Gilsonite (an asphaltite) and sometimes coal are used as extremely low-specific-gravity solids for reducing slurry density without greatly increasing the water content. The addition of gilsonite has almost no effect on slurry thickening time, and low-density cements obtained using gilsonite have much higher compressive strengths than other types of low-density cements. Example properties obtained with gilsonite and Class A cements are shown in Table 3.12.

3.4.5 Expanded Perlite. Perlite is volcanic glass containing a small amount of combined water. The raw ore is expanded by introduction into a kiln, where the temperature is raised to the fusion point of

TABLE 3.10—CLASS A CEMENT WITH BENTONITE⁴

Bentonite (%)	Maximum Water Requirements		Slurry Weight		Slurry Volume (cu ft/sk)
	(gal/sk)	(cu ft/sk)	(lbm/gal)	(lbm/cu ft)	
0	5.2	0.70	15.6	117	1.18
2	6.5	0.87	14.7	110	1.36
4	7.8	1.04	14.1	105	1.55
6	9.1	1.22	13.5	101	1.73
8	10.4	1.39	13.1	98	1.92

Thickening Time—Hours: Minutes
(Pressure-Temperature Thickening-Time Test)

Bentonite (%)	API Casing Tests			API Squeeze Tests		
	4,000 ft	6,000 ft	8,000 ft	2,000 ft	4,000 ft	6,000 ft
0	3:00 +	2:25	1:40	2:14	1:32	1:01
2	2:25	1:48	1:34	2:25	1:29	0:56
4	2:34	1:57	1:32	2:26	1:18	0:58
6	2:35	1:45	1:22	2:16	1:26	0:56
8	2:44	1:50	1:24	2:31	1:28	0:58

Compressive Strengths—psi
Atmospheric Pressure

Bentonite (%)					Bentonite (%)				
	60°F	80°F	100°F	120°F		60°F	80°F	100°F	120°F
12 Hours					24 Hours				
0	80	580	1,035	1,905	0	615	1,905	2,610	3,595
2	55	455	635	1,280	2	365	1,090	1,520	2,040
4	20	220	375	780	4	225	750	1,015	1,380
6	15	85	245	500	6	85	360	730	925
8	15	50	155	310	8	60	265	510	610

TABLE 3.11 – CLASS A CEMENT WITH
DIATOMACEOUS EARTH⁴

Diacel D (%)	Water (gal/sk)	Slurry Weight (lbm/gal)	Slurry Volume (cu ft/sk)
0	5.2	15.6	1.18
10	10.2	13.2	1.92
20	13.5	12.4	2.42
30	18.2	11.7	3.12
40	25.6	11.0	4.19

Thickening Time – Hours:Minutes
(Pressure-Temperature Thickening-Time Test)

Diacel D (%)	API Casing Tests		
	4,000 ft	6,000 ft	8,000 ft
0	3:36	2:41	1:59
10	4:00 +	3:00 +	2:14
20	4:00 +	3:00 +	2:38
30	4:00 +	3:00 +	3:00 +
40	4:00 +	3:00 +	3:00 +

Compressive Strength (psi)

Diacel D (%)	Temperature			
	80°F	100°F	120°F	140°F
24 Hours				
0	875	2,305	3,850	4,690
10	280	620	1,125	1,435
20	140	295	660	1,215
40	20	45	190	635
72 Hours				
0	3,535	5,965	7,285	7,090
10	1,010	1,430	2,140	2,210
20	530	905	1,895	1,860
40	95	245	820	795

TABLE 3.12 – CLASS A CEMENT WITH GILSONITE³

Gilsonite (lbm/sk)	Water Ratio		Slurry Weight		Slurry Weight (cu ft/sk)
	(gal/sk)	(cu ft/sk)	(lbm/gal)	(lbm/cu ft)	
0	5.2	0.70	15.6	117	1.18
5	5.4	0.72	15.1	113	1.27
10	5.4	0.72	14.7	110	1.36
12.5	5.6	0.75	14.4	108	1.42
15	5.7	0.76	14.3	107	1.47
20	5.7	0.76	14.0	105	1.55
25	6.0	0.80	13.6	102	1.66
50	7.0	0.94	12.5	94	2.17
100	9.0	1.20	11.3	85	3.18

the ore. At this temperature, the combined water in the ore expands, producing a cellular, thin-walled structure. Mixing water will enter this cellular structure under high pressure when perlite is used in cements. Laboratory tests have shown that approximately 4.5 gallons of water are required to completely saturate 1 cu ft of expanded perlite under pressure. If this additional water is not included in the slurry, the loss of slurry water to the perlite will cause the slurry to become too viscous to pump.

Expanded perlite is marketed in a variety of blends. The term “regular” usually is applied to the unblended material. Blends of 13 lbm expanded perlite mixed with 30 lbm of waste volcanic glass fines and blends of 8 lbm of expanded perlite with 30 lbm of pozzolanic material also are common.

TABLE 3.13 – CLASS A CEMENT WITH EXPANDED PERLITE³

		Atmospheric Pressure		Pressure, 3,000 psi	
Bentonite (%)	Water (gal/sk)	Slurry Weight (lbm/gal)	Slurry Volume (cu ft/sk)	Slurry Weight (lbm/gal)	Slurry Volume (cu ft/sk)
1 Sack Cement, 1/2 cu ft Oil Patch Regular					
2	8.5	13.01	1.78	13.84	1.68
4	9.7	12.70	1.95	13.43	1.84
6	10.8	12.44	2.11	13.10	2.00
8	11.7	12.29	2.24	12.90	2.14
1 Sack Cement, 1 cu ft Oil Patch Regular					
2	10.5	11.91	2.20	13.20	1.99
4	11.7	11.74	2.28	12.90	2.16
6	12.8	11.60	2.54	12.83	2.32
8	13.7	11.50	2.66	12.50	2.45

TABLE 3.14 – CLASS A CEMENT WITH POZZOLAN³

Mixture (%)		Water Weight (lbm/sk)	Water Solids Ratio	Water Ratio		Slurry Weight		Slurry Volume (cu ft/sk)
Pozmix "A"	Portland Cement			gals/sk of Mix	cu ft/sk of Mix	(lbm/gal)	(lbm cu ft)	
0	100	94	0.46	5.20	0.70	15.60	117	1.18
25	75	89	0.56	5.98	0.80	14.55	109	1.29
*50	50	84	0.57	5.75	0.77	14.15	106	1.26
60	40	82	0.58	5.71	0.76	14.00	105	1.25
75	25	79	0.63	5.97	0.80	13.55	101	1.29

*Most commonly recommended blend.

TABLE 3.15 – CLASS E CEMENT WITH HEMATITE³

Hematite Hi-Dense No. 3 (lbm/sk cement)	Water (gal/sk)	Slurry Volume (cu ft/sk)	Slurry Weight (lbm/gal)	24-Hour Compressive Strength (psi), Curing Pressure 3,000 psi	
				260°F	290°F
0	4.5	1.08	16.25	6,965	4,125
12	4.55	1.12	17.0	6,425	4,090
28	4.6	1.18	18.0	6,290	4,275
46	4.65	1.24	19.0	5,915	5,575

Bentonite generally is used with expanded perlite to minimize water separation at the surface and to raise the viscosity. Without bentonite, the perlite tends to separate and float above the slurry because of its low density. The approximate water requirements for various concentrations of perlite and water are shown in Table 3.13. Also shown are the slurry densities obtained at atmospheric pressure and at a pressure of 3,000 psi. The density increase at the elevated pressure results from water in the slurry being forced into the cellular structure of the expanded perlite.

3.4.6 Pozzolan. Pozzolans are siliceous and aluminous mineral substances that will react with calcium hydroxide formed in the hydration of portland cement to form calcium silicates that possess cementitious properties. Diatomaceous earth, which has been discussed previously, is an example of a pozzolan. However, the term pozzolan as used in marketing cement additives usually refers

to finely ground pumice or fly ash (flue dust) produced in coal-burning power plants. The specific gravity of pozzolans is only slightly less than the specific gravity of portland cement, and the water requirement of pozzolans is about the same as for portland cements. Thus, only slight reductions in density can be achieved with this material. The range of slurry densities possible using various concentrations of one type of pozzolan is shown in Table 3.14. Because of this relatively low cost, considerable cost savings can be achieved through the use of pozzolans.

3.4.7 Hematite. Hematite is reddish iron oxide ore (Fe_2O_3) having a specific gravity of approximately 5.02. Hematite can be used to increase the density of a cement slurry to as high as 19 lbm/gal. Metallic powders having a higher specific gravity than hematite have been tried but were found to settle out of the slurry rapidly unless they were ground extremely fine. When ground fine enough to prevent settling, the

TABLE 3.16 – CLASS A, B, G, or H CEMENT WITH ILMENITE³

Ilmenite Hi-Dense No. 2 (lbm/sk cement)	Water (gal/sk)	Slurry Volume (cu ft/sk)	Slurry Weight (lbm/gal)
0	5.2	1.18	15.6
7	5.2	1.20	16.0
22	5.2	1.25	17.0
39	5.2	1.31	18.0

TABLE 3.17 – CLASS E CEMENT WITH BARITE³

Barite (lbm/sk)	Water (gal/sk)	Slurry Volume (cu ft/sk)	Slurry Weight (lbm/gal)	Thickening Time, 14,000 ft. Casing Schedule (hours:minutes)	24-Hour Compressive Strength (psi) Curing Pressure, 3,000 psi 260 F 290 F
0	4.5	1.08	16.25	3:00 +	6.965 4.125
22	5.1	1.24	17.0	3:00 +	4.440 4.225
55	5.8	1.46	18.0	2:28	4.315 4.000
108	7.1	1.83	19.0	1:45	3.515 2.650

TABLE 3.18 – RETARDED CEMENT WITH OTTAWA SAND³

Ottawa Sand (20-40) (lbm/sk cement)	Water (gal/sk)	Slurry Volume (cu ft/sk)	Slurry Weight (lbm/gal)
0	4.5	1.08	16.25
10	4.5	1.14	16.50
28	4.5	1.25	17.00
51	4.5	1.39	17.50
79	4.5	1.56	18.00

increased water requirement results in slurry densities below that possible with hematite. The water requirement for hematite is approximately 0.36 gal/100 lbm hematite. The effect of hematite on the thickening time and compressive strength of the cement has been found to be minimal at the concentrations of hematite generally used. The range of slurry densities possible using various concentrations of hematite is shown in Table 3.15.

3.4.8 Ilmenite. Ilmenite is a black mineral composed of iron, titanium, and oxygen that has a specific gravity of approximately 4.67. Although ilmenite has a slightly lower specific gravity than hematite, it requires no additional water and provides about the same slurry density increase as hematite at comparable concentrations. Like hematite, ilmenite has little effect on thickening time or compressive strength. The range of slurry densities possible using various concentrations of ilmenite is shown in Table 3.16.

3.4.9 Barite. The use of barite, or barium sulfate, for increasing the density of drilling fluids has been discussed previously in Chap. 2. This mineral also is used extensively for increasing the density of a cement slurry. The water requirements for barite are considerably higher than for hematite or ilmenite,

TABLE 3.19 – CLASS A CEMENT WITH CALCIUM CHLORIDE⁴

Bentonite (%)	Water Requirement (gal/sk)	Slurry Weight (lbm/gal)	Slurry Volume (cu ft/sk)
0	5.2	15.6	1.18
2	6.5	14.7	1.36
4	7.8	14.1	1.55

Thickening Time (hours:minutes)
(pressure-temperature thickening-time test)

API Casing Tests

Calcium Chloride (%)	0% Bentonite		2% Bentonite		4% Bentonite	
	2,000 ft	4,000 ft	2,000 ft	4,000 ft	2,000 ft	4,000 ft
0	3:36	2:25	3:20	2:25	3:46	2:34
2	1:30	1:04	2:00	1:30	2:41	2:03
4	0:47	0:41	0:56	1:10	1:52	2:00

API Compressive Strength (psi)

0% Bentonite

Calcium Chloride (%)	Curing Temperature and Pressure				
	60°F, 0 psi	80°F, 0 psi	95°F, 800 psi	110°F, 1,600 psi	140°F, 3,000 psi

6 Hours

0	Not Set	115	260	585	2,060
2	190	685	945	1,220	2,680
4	355	960	1,195	1,600	3,060

8 Hours

0	20	265	445	730	2,890
2	300	1,230	1,250	1,750	3,380
4	450	1,490	1,650	2,350	2,950

12 Hours

0	80	580	800	1,120	3,170
2	555	1,675	2,310	2,680	3,545
4	705	2,010	2,500	3,725	4,060

18 Hours

0	375	1,405	1,725	2,525	3,890
2	970	2,520	3,000	4,140	5,890
4	1,105	2,800	3,640	4,690	4,850

24 Hours

0	615	1,905	2,085	2,925	5,050
2	1,450	3,125	3,750	5,015	6,110
4	1,695	3,080	4,375	4,600	5,410

requiring about 2.4 gal/100 lbm of barite. The large amount of water required decreases the compressive strength of the cement and dilutes the other chemical additives. The range of slurry densities possible using various concentrations of barite is shown in Table 3.17.

3.4.10 Sand. Ottawa sand, even though it has a relatively low specific gravity of about 2.63, sometimes is used to increase slurry density. This is possible since the sand requires no additional water to be added to the slurry. Sand has little effect on the strength or pumpability of the cement, but causes the cement surface to be relatively hard. Because of the tendency to form a hard cement, sand often is used to form a plug in an open hole as a base for setting a whipstock tool used to change the direction of the hole. The range of slurry densities possible using

various concentrations of sand is shown in Table 3.18.

Example 3.5. It is desired to increase the density of a Class H cement slurry to 17.5 lbm/gal. Compute the amount of hematite that should be blended with each sack of cement. The water requirements are 4.5 gal/94 lbm Class H cement and 0.36 gal/100 lbm hematite.

Solution. Let x represent the pounds of hematite per sack of cement. The total water requirement of the slurry then is given by $4.5 + 0.0036x$. Expressing the slurry density in terms of x yields

$$\rho = \frac{\text{total mass, lbm}}{\text{total volume, gal}}$$

$$17.5 = \frac{94 + x + 8.34(4.5 + 0.0036x)}{\left[\frac{94}{3.14(8.34)} + \frac{x}{5.02(8.34)} + (4.5 + 0.0036x) \right]}$$

Solving this expression yields $x = 18.3$ lbm hematite/94 lbm cement.

3.4.11 Setting-Time Control. The cement must set and develop sufficient strength to support the casing and seal off fluid movement behind the casing before drilling or completion activities can be resumed. The exact amounts of compressive strength needed is difficult to determine, but a value of 500 psi commonly is used in field practice. Experimental work by Farris⁶ has shown that a tensile strength of only a few psi was sufficient to support the weight of the casing under laboratory conditions. However, some consideration also must be given to the shock loading imposed by the rotating drillstring during subsequent drilling operations. It is possible for the drillstring to knock off the lower joint of casing and junk the hole if a good bond is not obtained. The cement strength required to prevent significant fluid movement behind the casing was investigated by Clark.⁷ His data show that tensile strengths as low as 40 psi are acceptable with maximum bonding being reached at a value of about 100 psi. Since the ratio of compressive strength to tensile strengths usually is about 12:1, 40- and 100-psi tensile strengths correspond to compressive strengths of 480 and 1,200 psi.

When cementing shallow, low-temperature wells, it may be necessary to accelerate the cement hydration so that the waiting period after cementing is minimized. The commonly used cement accelerators are (1) calcium chloride, (2) sodium chloride, (3) hemihydrate form of gypsum, and (4) sodium silicate. Cement setting time also is a function of the cement composition, fineness, and water content. For example, API Class C cement is ground finer and has a higher C_3A content to promote rapid hydration. When low water/cement ratios are used to reduce setting time, friction-reducing agents (dispersants) sometimes are used to control rheological properties. However, the dispersant must be chosen

with care since many dispersants tend to retard the setting of the cement. Organic dispersants such as tannins and lignins already may be present in the water available for mixing cement, especially in swampy locations. Thus, it often is important to measure cement thickening time using a water sample taken from the location.

3.4.12 Calcium Chloride. Calcium chloride in concentrations up to 4% by weight commonly is used as a cement accelerator in wells having bottomhole temperatures of less than 125°F. It is available in a regular grade (77% calcium chloride) and an anhydrous grade (96% calcium chloride). The anhydrous grade is in more general use because it absorbs moisture less readily and is easier to maintain in storage. The effect of calcium chloride on the compressive strength of API Class A cement is shown in Table 3.19.

3.4.13 Sodium Chloride. Sodium chloride is an accelerator when used in low concentrations. Maximum acceleration occurs at a concentration of about 5% (by weight of mixing water) for cements containing no bentonite. At concentrations above 5%, the effectiveness of sodium chloride as an accelerator is reduced. Saturated sodium chloride solutions tend to act as a retarder rather than an accelerator. Saturated sodium chloride cements are used primarily for cementing through salt formations and through shale formations that are highly sensitive to fresh water. Potassium chloride is more effective than sodium chloride for inhibiting shale hydration and can be used for this purpose when the additional cost is justified. The effect of sodium chloride on the compressive strength of API Class A cement is shown in Table 3.20.

Seawater often is used for mixing cement when drilling offshore. The sodium, magnesium, and calcium chlorides at the concentrations present in the seawater all act as cement accelerators. Typical effects of seawater on cement slurry properties as compared with fresh water are shown in Table 3.21.

This thickening time obtained with seawater usually is adequate for cement placement where bottomhole temperatures do not exceed 160°F. Cement retarders can be used to counteract the effect of the seawater at higher temperatures, but laboratory tests always should be made before this type of application.

3.4.14 Gypsum. Special grades of gypsum hemihydrate cement can be blended with portland cement to produce a cement with a low thickening time at low temperatures. These materials should not be used at high temperatures, because the gypsum hydrates may not form a stable set. The maximum working temperature depends on the grade of gypsum cement used, varying from 140°F for the regular grade to 180°F for the high-temperature grade. A full range of blends, from as little as 1 sack gypsum/20 sacks cement to pure gypsum, have been used for various applications. The water requirement of gypsum hemihydrate is about 4.8 gal/100-lbm sack.

TABLE 3.20—CLASS A CEMENT WITH SODIUM CHLORIDE⁴

Water Requirements		% Salt by Weight of Water	Weight of Dry Salt (lbm/sk cement)	Slurry Weight		Slurry Volume (cu ft/sk)
(gal/sk)	(cu ft/sk)			(lbm/gal)	(lbm/cu ft)	
5.2	0.70	0	0	15.6	117	1.18
		5	2.17	15.7	117	1.19
		10	4.33	15.8	118	1.20
		15	6.50	15.9	119	1.21
		20	8.66	16.0	120	1.22
		sat (140°F)	16.12	16.1	120	1.27

Thickening Time and Compressive Strength

Salt (%)	Calcium Chloride (%)	Thickening Time, 2,000 ft, Casing Test (hours:minutes)	Compressive Strength (psi)			
			8 Hours		24 Hours	
			95 F, 800 psi	110 F, 1,600 psi	95 F, 800 psi	110 F, 1,600 psi
0	0	4:15	305	925	2,240	3,230
0	2	1:40	1,365	2,000	3,920	4,815
5	0	2:30	1,050	2,060	3,990	4,350
5	2	1:49	1,630	2,515	4,530	5,465
10	0	2:30	965	1,925	4,150	4,730
10	2	1:48	1,235	2,200	3,775	4,650
15	0	3:01	700	1,735	4,015	4,480
15	2	2:31	945	1,605	3,075	3,820
20	0	3:00	380	1,140	3,175	3,495
20	2	3:13	490	1,065	2,390	3,155
sat	0	7:15 +	not set	15	930	1,955
sat	2	5:00 +	50	290	1,570	2,450

API Class A Cement With HR-4 Retarder, Water, 5.2 gal/sk

Thickening Time (hours:minutes)

HR-4 (%)	API Casing Cementing				API Squeeze Cementing		
	4,000 ft	6,000 ft	8,000 ft	10,000 ft	4,000 ft	6,000 ft	8,000 ft
0% Salt							
0.0	3:36	2:25	1:59	1:14	1:32	1:01	0:44
10% Salt Water							
0.0	1:53	1:30	1:10	0:50	1:08	1:00	0:30
0.2	2:29	2:05	1:33	1:18	1:43	1:15	0:34
0.4	3:00 +	3:00 +	3:08	3:14	2:59	2:45	—
15% Salt Water							
0.0	2:05	1:33	1:25	1:10	1:35	0:55	0:21
0.2	3:00 +	2:17	2:02	1:48	2:19	2:07	0:22
0.4	3:00 +	3:00 +	3:00 +	3:00	—	—	0:15
20% Salt Water							
0.0	2:00 +	2:00	1:50	1:13	1:47	1:23	0:42
0.2	3:00 +	3:00 +	2:36	1:36	3:00 +	2:12	—

For very shallow wells and surface applications at low temperatures where an extremely short setting time combined with rapid strength development is desired, a small amount of sodium chloride can be used with a gypsum cement blend. For example, a laboratory blend of 90 lbm of gypsum hemihydrate, 10 lbm of Class A portland cement, and 2 lbm of salt when mixed with 4.8 gal of water will develop over 1,000 psi of compression strength when cured at only 50°F for 30 minutes.

3.4.15 Sodium Silicate. Sodium silicate is used as an accelerator for cements containing diatomaceous

earth. It is used in concentrations up to about 7% by weight.

3.4.16 Cement Retarders. Most of the organic compounds discussed in Chap. 2 for use as drilling fluid deflocculants tend to retard the setting of portland cement slurries. These materials also are called thinners or dispersants. Calcium lignosulfonate, one of the common mud deflocculants, has been found to be very effective as a cement retarder at very low concentrations. Laboratory data on the thickening time of Class A and Class H cements at various concentrations of calcium lignosulfonate are shown in Table 3.22.

TABLE 3.21 – TYPICAL EFFECT OF SEAWATER ON THICKENING TIME
(water ratio: 5.2 gal/sk)

	Thickening Time Hours:Minutes		Compressive Strength (psi at 24 hours)		
	6,000 ft*	8,000 ft*	50°F	110°F, 1,600 psi	140°F, 3,000 psi
API Class A Cement					
Fresh water	2:25	1:59	435	3,230	4,025
Seawater	1:33	1:17	520	4,105	4,670
API Class H Cement					
Fresh water	2:59	2:16	—	1,410	2,575
Seawater	1:47	1:20	—	2,500	3,085

*API RP 10B Casing Schedule.

The addition of an organic acid to the calcium lignosulfonate (Halliburton HR-12™) has been found to give excellent retarding characteristics at extremely high temperatures. It also improves the rheological properties of the slurry to a greater extent than calcium lignosulfonate alone. When the addition of the organic acid increases the effectiveness of the retarder to the extent that less than 0.3% would be used, it may be difficult to obtain a uniform blend. In this case, the use of calcium lignosulfonate is best. The addition of the organic acid also has been found to be effective in Class A cement.

Calcium-sodium lignosulfonate has been found to be superior to a calcium lignosulfonate when high concentrations of bentonite are used in the cement. The use of calcium-sodium lignosulfonate has been found to produce a slurry having a lower viscosity during mixing and helps to reduce air entrainment.

Sodium tetraborate decahydrate (borax) can be used to enhance the effectiveness of the organic deflocculants as retarders, especially in deep, high-temperature wells where a large increase in thickening time is needed. The optimum borax concentration is thought to be about one-third of the concentration of deflocculant used. Laboratory tests have shown that in addition to increasing the pumping time, the borax reduces the detrimental effect of the deflocculant on the early compressive strength of the cement.

Carboxymethyl hydroxyethyl cellulose (CMHEC) commonly is used both for cement retardation and for fluid loss control. It is used more commonly with cements containing diatomaceous earth, but it is effective as a retarder in essentially all portland cements. Laboratory data on the thickening time and compressive strength of Class H cement at various concentrations of CMHEC are shown in Table 3.23.

**TABLE 3.22 – CLASS A AND CLASS H CEMENT
WITH CALCIUM LIGNOSULFONATE
(Halliburton HR-4)³**

API Class A Cement or Pozmix Cement

Depth (ft)	Temperature (°F)		Calcium Lignosulfate	Approximate Thickening Time (hours)
	Static	Circulating		
Casing Cementing (Primary)				
2,000 to 6,000	110 to 170	91 to 113	0.0	2 to 4
6,000 to 10,000	170 to 230	113 to 144	0.0 to 0.5	2 to 4
10,000 to 14,000	230 to 290	144 to 206	0.5 to 1.0	3 to 4
Squeeze Cementing				
2,000 to 4,000	110 to 140	98 to 116	0.0	2 to 4
4,000 to 8,000	140 to 200	116 to 159	0.0 to 0.5	2 to 4
8,000 to 12,000	200 to 260	159 to 213	0.5 to 1.0	3 to 4

API Class H Cement

Casing Cementing				
4,000 to 6,000	140 to 170	103 to 113	0.0	3 to 4
6,000 to 10,000	170 to 230	113 to 144	0.0 to 0.3	3 to 4
10,000 to 14,000	230 to 290	144 to 206	0.3 to 0.6	2 to 4
14,000 to 18,000	290 to 350	206 to 300	0.6 to 1.0	—
Squeeze Cementing				
2,000 to 4,000	110 to 140	98 to 116	0.0	3 to 4
4,000 to 6,000	140 to 170	116 to 136	0.0 to 0.3	2 to 4
6,000 to 10,000	170 to 230	136 to 186	0.3 to 0.5	3 to 4
10,000 to 14,000	230 to 290	186 to 242	0.5 to 1.0	2 to 4

3.4.17 Lost-Circulation Additives. *Lost circulation* is defined as the loss of drilling fluid or cement from the well to subsurface formations. This condition is detected at the surface when the flow rate out of the annulus is less than the pump rate into the well. Lost circulation occurs when (1) extremely high-permeability formations are encountered, such as a gravel bed, oyster bed, or vugular limestone, or (2) a fractured formation is encountered or created because of excessive wellbore pressure.

Lost circulation usually occurs while drilling and can be overcome by adding lost-circulation material to the drilling fluid or reducing the drilling fluid density. In some cases, however, lost-circulation material is added to the cement slurry to minimize the loss of cement to a troublesome formation during cementing and, thus, to ensure placing the cement in the desired location. The lost-circulation additives are classified as (1) fibrous, (2) granular, or (3) lamellated. In laboratory experiments, fibrous and granular additives are effective in high-permeability gravel beds. In simulated fractures, granular and lamellated additives are found to be effective. The commonly used granular additives include gilsonite, expanded perlite, plastics, and crushed walnut shells. Fibrous materials used include nylon fibers, shredded wood bark, sawdust, and hay. However, the use of wood products can cause cement retardation because they contain tannins. Lamellated materials include cellophane and mica flakes.

Semi-solid and flash-setting slurries are available for stopping severe lost-circulation problems encountered while drilling that cannot be remedied by adding lost-circulation additives to the mud. The placement of these cements in the lost-circulation zone requires a special operation, since they are not merely additives to the fluid being circulated while drilling. The slurries most often used for this purpose include (1) the gypsum hemihydrate cements, (2) mixtures of bentonite and diesel oil (gunk), and (3) mixtures of cement, bentonite, and diesel oil (bengum).

3.4.18 Filtration-Control Additives. Cement filtration-control additives serve the same function as the mud filtration-control additives discussed in Chap. 2. However, cement slurries containing no filtration control additives have much higher filtration rates than clay/water muds. An untreated slurry of Class H cement has a 30-minute API filter loss in excess of 1,000 cm³. It is desirable to limit the loss of water filtrate from the slurry to permeable formations to (1) minimize the hydration of formations containing water-sensitive shales, (2) prevent increases in slurry viscosity during cement placement, (3) prevent the formation of annular bridges, which can act as a packer and remove hydrostatic pressure holding back potentially dangerous high-pressure zones, and (4) reduce the rate of cement dehydration while pumping cement into abandoned perforated intervals and, thus, allow plugging longer perforated intervals in a single operation.

The commonly used filtration-control additives include (1) latex, (2) bentonite with a dispersant, (3) CMHEC, and (4) various organic polymers. The

TABLE 3.23—CLASS H CEMENT WITH CMHEC⁴

Water Requirement		Slurry Density		Slurry Volume (cu ft/sk)
(gal/sk)	(cu ft/sk)	(lbm/gal)	(lbm/cu ft)	
4.3	0.58	16.4	123	1.06
5.2	0.70	15.6	117	1.18

Thickening Time (hours:minutes)
(Pressure-Temperature Thickening-Time Tests)

Diacel LWL %	Well Depth (ft), Water, 4.3 gal/sk		
	10,000	12,000	14,000
0.05	1:34	1:28	0:53
0.10	3:52	2:24	2:07
0.15	4:00 +	3:37	2:24
0.20	4:00 +	4:00 +	3:04

Compressive Strength (psi)

Diacel LWL (%)	Curing Time (hours)	Curing Temperature (°F, 3,000 psi), Water, 4.3 gal/sk			
		200	230	260	290
0.00	8	5,450	6,350	6,950	6,375
	24	8,400	7,950	8,525	7,700
0.05	8	3,500	5,850	6,850	6,750
	24	8,350	8,125	8,800	7,200
0.10	8	2,025	5,100	6,700	7,050
	24	8,200	8,300	9,000	6,700
0.15	8	850	3,600	6,400	7,250
	24	7,200	8,450	9,150	6,200
0.20	8	0	1,750	6,100	7,400
	24	4,925	8,600	9,200	5,650

Thickening Time (hours:minutes)
(Pressure-Temperature Thickening-Time Tests)

Diacel LWL %	Well Depth (ft), Water, 5.2 gal/sk		
	10,000	12,000	14,000
0.05	2:32	1:40	1:04
0.10	4:12	3:28	2:30
0.15	4:00 +	4:23	2:35
0.20	4:00 +	4:00 +	3:32

filtration rate in cm³/30 min through a 325-mesh screen caused by a 1,000-psi pressure differential for various concentrations of one of the organic polymers (Halliburton HALAD-9TM) in Class H cement is shown in Table 3.24.

3.4.19 Viscosity-Control Additives. Untreated cement slurries have a high effective viscosity at the shear rates present during cement placement. It is desirable to reduce the effective viscosity of the slurry so that (1) less pump horsepower will be required for cement placement, (2) there will be reduction in the annular frictional pressure gradient and, thus, a smaller chance of formation fracture, and (3) the slurry can be placed in turbulent flow at a lower pumping rate. Some evidence indicates that the drilling fluid is displaced with less mixing and, thus, less cement contamination when the flow pattern is turbulent. Shown in Table 3.25 is a comparison of the velocity required to achieve turbulence of a Class H cement with and without a viscosity-control additive. The commonly used viscosity-control additives include (1) the organic deflocculants such as

TABLE 3.24 – CLASS H CEMENT WITH A FILTRATION CONTROL ADDITIVE³

HALAD-9 (%)	Salt (%)	Water Requirement		Slurry Weight		Slurry Volume (cu ft/sk)
		(gal/sk)	(cu ft/sk)	(lbm/gal)	(lbm/cu ft)	
0.0	0	5.20	0.70	15.60	117	1.18
0.6 to 1.2	0	5.64	0.75	15.26	114	1.24

Fluid Loss Tests
(cm³/30 min; 325-mesh screen; 1,000-psi pressure)

HALAD-9 (%)	0% Salt	10% Salt
0.6	72	192
0.8	52	84
1.0	38	62
1.2	24	36

calcium lignosulfonate, (2) sodium chloride, and (3) certain long-chain polymers. Deflocculants reduce cement viscosity in the same manner as discussed previously in Chap. 2 for drilling fluids. However, it should be remembered that deflocculants act as retarders as well as thinners. Certain organic polymers are available that will act as thinners without accelerating or retarding the cement.

3.4.20 Other Additives. Miscellaneous additives and slurries not discussed in the previous categories given include (1) paraformaldehyde and sodium chromate, which are used to counteract the effect of cement contamination by organic deflocculants from the drilling mud, (2) silica flour, which is used to form a stronger, more stable, and less permeable cement for high-temperature applications, (3) hydrazine, an oxygen scavenger used to control corrosion, (4) radioactive tracers used to determine where the cement has been placed, (5) special fibers such as nylon to make the cement more impact resistant, and (6) special compounds which slowly evolve small gas bubbles as the cement begins to harden.

The formation of small gas bubbles in the cement is thought to be desirable when there is danger of gas flow occurring in the newly cemented borehole when the cement begins to harden. There have been several

cases of gas flow through the annulus several hours after cementing operations were completed. A gas flow outside the casing can be particularly difficult to stop because conventional well control procedures cannot be used easily.

The mechanism by which gas blowouts occur shortly after cementing is not fully understood. However, it is known that the formation of a semirigid gel structure begins as soon as cement placement is completed. Initially, the formation of the static gel structure is similar to that occurring in a drilling fluid when fluid movement is stopped. Later, as the cement begins to set, the cement gel becomes much more progressive than that of a drilling fluid. As the cement slurry goes through a transition from a liquid to a solid, it begins to lose the ability to transmit hydrostatic pressure to the lower part of the cemented annulus. (Equations applicable to this phenomenon are developed later in Sec. 12 of Chap. 4.) If this loss in ability to transmit hydrostatic pressure is accompanied by a cement slurry volume reduction, the wellbore pressure can fall sufficiently to permit gas from a permeable high-pressure formation to enter the annulus. The semirigid slurry may not be able to withstand the higher stresses created when gas begins to flow. Gas flow may increase and communicate with a more shallow formation. In an extreme case, gas flow may reach the surface.

TABLE 3.25 – CRITICAL FLOW RATES FOR TURBULENCE³

API Class H Cement, Water Ratio: 5.2 gal/sk Slurry Weight: 15.6 lbm/gal					
Slurry Properties		Neat	Dispersant, 1.0%		
n', flow behavior		0.30	0.67		
K', consistency index		0.195	0.004		
Pipe Size (in.)	Hole Size (in.)	Critical Velocity (ft/sec)		Critical Flow Rate (bbl/min)	
		Without Dispersant	With Dispersant	Without Dispersant	With Dispersant
4½	6¾	9.2	2.61	13.6	3.85
4½	7⅞	8.6	2.13	20.9	5.18
5½	7⅞	9.1	2.54	16.9	4.70
5½	8¾	8.6	2.17	23.3	5.85
7	8¾	9.6	2.96	15.5	4.76
8⅝	11	9.1	2.54	24.4	6.90
8⅝	12¼	8.5	2.05	37.0	9.05
9⅝	12¼	9.0	2.41	29.6	8.08

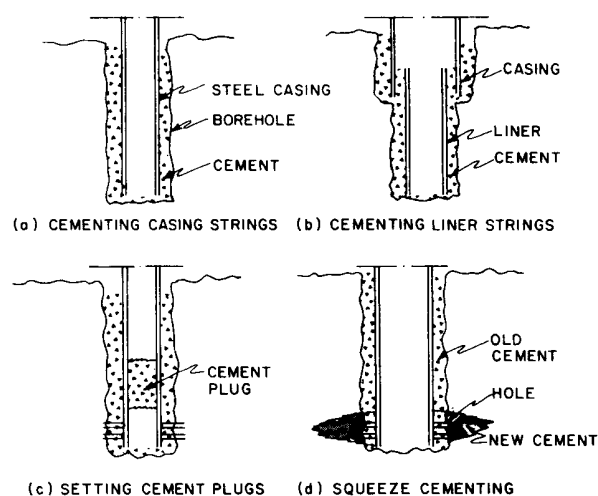


Fig. 3.6 – Common cement placement requirements.

Volume reductions occurring while the cement is making the transition from a liquid slurry to a rigid solid can be traced to two sources. A small volume reduction, as measured in the soundness test, occurs due to the cement hydration reaction. For most cements used in current practice, this volume reduction is small, generally being on the order of 0.1 to 0.3%.¹⁰ Much larger volume reductions are thought to be possible due to the loss of water filtrate to the borehole walls.

The magnitude of the pressure loss per unit volume of filtrate loss is controlled primarily by the cement compressibility during the early stages of the hardening process. A cement with a high compressibility is desirable because it will give a small pressure loss per unit volume of filtrate loss. The introduction of a compound that will react slowly to form small gas bubbles as the cement begins to harden will greatly increase the compressibility of the cement. Cement compressibility also can be increased by blending small volumes of nitrogen gas with the cement slurry during mixing.

A high-compressibility cement permits much larger volumes of water filtrate to be lost without greatly increasing the potential for gas flow into the well. Other methods for reducing the potential for gas flow after cementing include use of (1) filtration control additives to reduce the volume of filtrate loss, (2) shorter cement columns to reduce the effectiveness of the gel strength in blocking the transmission of hydrostatic pressure, and (3) cements that build gel strength quickly after pumping is stopped and harden more rapidly.

3.5. Cement Placement Techniques

Different cementing equipment and placement techniques are used for (1) cementing *casing* strings, (2) cementing *liner* strings, (3) setting cement *plugs*, and (4) *squeeze* cementing. These different types of cementing applications are illustrated in Fig. 3.6. A

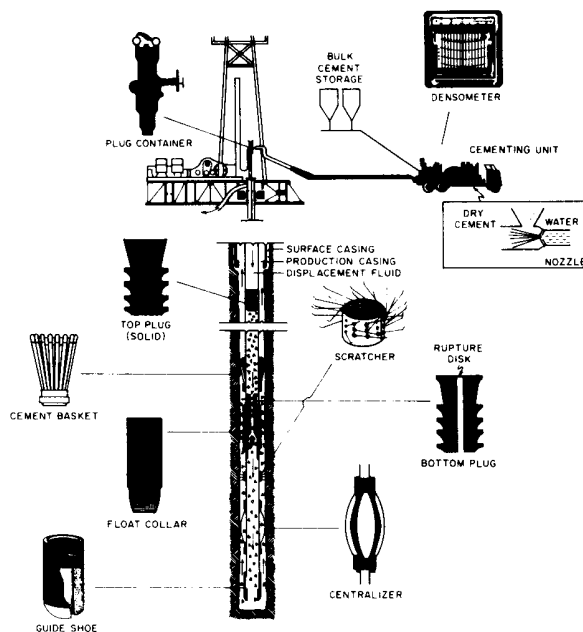


Fig. 3.7 – Conventional placement technique used for cementing casing.⁵

casing string differs from a liner in that casing extends to the surface, while the top of a liner is attached to subsurface casing previously cemented in place. Cement plugs are placed in open hole or in casing before abandoning the lower portion of the well. Cement is squeezed into lost-circulation zones, abandoned casing perforations, or a leaking cemented zone to stop undesired fluid movement.

3.5.1 Cement Casing. The conventional method for cementing casing is described in Fig. 3.7. Cement of the desired composition is blended at a bulk blending station where cement is moved by aerating the fine powder and blowing it between pressurized vessels under 30 to 40 psi air pressure. The blended cement is transported to the job in bulk transport units. When the casing string is ready to be cemented, cement is mixed with water in a special cementing unit. The cementing unit usually is truck-mounted for land jobs and skid-mounted for offshore operations. The unit mixes the slurry by pumping water under high pressure through a nozzle and loading dry cement through a hopper just downstream of the nozzle. Dry cement is moved pneumatically from the bulk units. The cement slurry enters a pump on the cementing unit and is pumped to a special cementing head or plug container screwed into the top joint of casing.

When the cementing operation begins, the bottom rubber wiper plug is released from the plug container ahead of the cement slurry. This plug wipes the mud from the casing ahead of the slurry to minimize the contamination of the cement with the mud. In some cases a slug of liquid called a *mud preflush* is pumped into the casing before cementing to assist in isolating the cement from the drilling fluid. After the desired volume of slurry has been mixed and pumped into the casing, a top wiper plug is released from the plug container. The top plug differs from the bottom plug

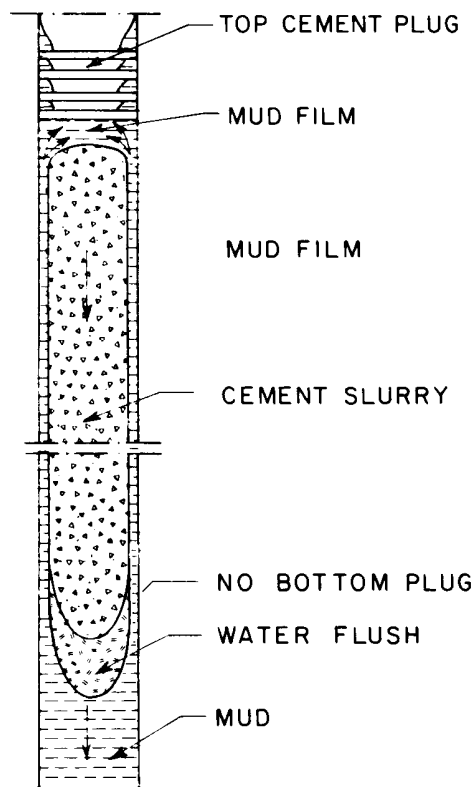
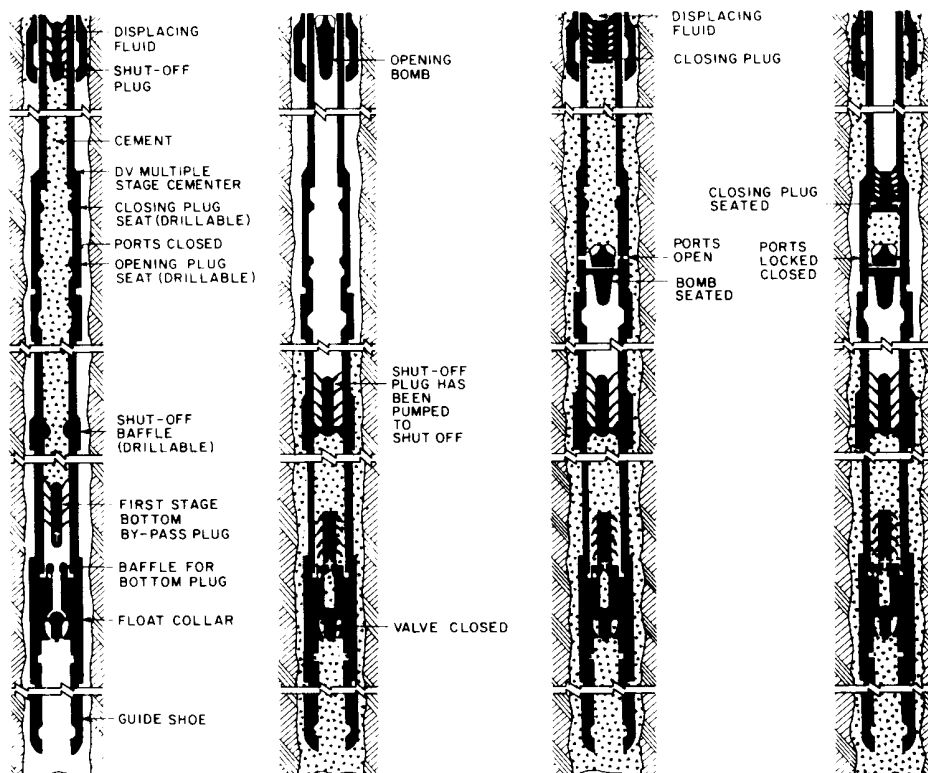


Fig. 3.8 – Cement contamination without bottom plug.

in that the top plug is solid rubber while the bottom plug contains a thin rupture diaphragm. The cement slug is displaced down the casing by pumping drilling fluid or completion fluid into the casing behind the top plug. When the bottom plug reaches the float collar, the diaphragm in the plug ruptures, allowing the cement slurry to be displaced through the guide shoe and into the annulus. When the top plug reaches the bottom plug, the pressure increase at the surface signifies the end of the displacement operation. By counting pump strokes, the approximate position of the top plug can be determined at all times and the pump can be slowed down at the end of the displacement to prevent an excessive pressure increase.

The float collar can act as a check valve to prevent cement from backing up into the casing. Top plugs that have pressure seals and latch in place can be used in addition to the float collar. Fluid movement also can be prevented by holding pressure on the top plug with the displacing fluid. However, it is not good practice to hold excessive pressure in the casing while the cement sets because when the pressure is released, the casing may change diameter sufficiently to break the bond with the cement and form a small annular channel.

Not all operators use a bottom plug when cementing. Indeed, there have been cases when the solid plug mistakenly was placed below the cement slurry. Also, with some of the first plug designs, the float collar was stopped up by plug fragments before completing the cement displacement. When a bottom plug is not used, however, the cement does not wipe all

Fig. 3.9 – Two-stage cementing.⁵

the mud from the wall of the casing. This results in a contaminated zone being built up in front of the top plug as shown in Fig. 3.8.

Subsurface equipment that often is used in the conventional casing cementing operation includes one or more joints of casing below the float collar, called the *shoe joints*, a *guide shoe* or *float shoe* at the bottom of the casing, and *centralizers*, *scratchers*, and *baskets* on the outside of the casing. The shoe joints allow for entrapment of contaminated mud or cement, which may result from the wiping action of the top cementing plug. The simplest guide shoe design is the open-ended collar type with or without a molded nose. The guide shoe simply guides the casing past irregularities in the borehole wall. Circulation is established through the open end of the guide shoe or through side ports designed to create more agitation as the cement slurry is circulated up the annulus. Should the casing be resting on bottom, circulation can be achieved more easily through side port openings in the guide shoe. A float shoe serves the function of both a guide shoe and a float collar when no shoe joints are desired. Float shoes and float collars containing a packer also are available for isolating the lower portion of the hole from the cemented zone. These devices have side ports for slurry exit above the packer. Centralizers are placed on the outside of the casing to help hold the casing in the center of the hole. Scratchers are used to help remove mudcake from the borehole walls. Some scratchers are designed for cleaning by reciprocating the casing, while others are designed for cleaning by rotating the casing. Cement baskets are used to help

support the weight of the cement slurry at points where porous or weak formations are exposed.

In addition to the conventional placement method for cementing casings, there are several modified techniques used in special situations. These include: (1) stage cementing, (2) inner-string cementing, (3) annular cementing through tubing, (4) multiple-string cementing, (5) reverse-circulation cementing, and (6) delayed-setting cementing.

3.5.2 Stage Cementing. Stage cementing is one of the procedures developed to permit using a cement column height in the annulus that normally would cause fracture of one or more subsurface formations. It also can be used to reduce the potential for gas flow after cementing. The first stage of the cementing operation is conducted in the conventional manner. After the slurry hardens, a bomb is dropped (Fig. 3.9) to open a side port in a staging tool placed in the casing string. The second-stage cement then is pumped through this side port and into the annulus above the set first-stage cement. Equipment also is available for three-stage cementing.

3.5.3 Inner-String Cementing. Inner-string cementing was developed to reduce the cementing time and the amount of cement left in the shoe joint of extremely large-diameter casing. The technique uses a float collar or shoe modified with sealing adapters, which permits tubing or drillpipe to be landed and hydraulically sealed. The cement then is displaced down the inner drillpipe or tubing string rather than the casing. When the float collar or shoe is equipped with a backpressure valve or latch-down

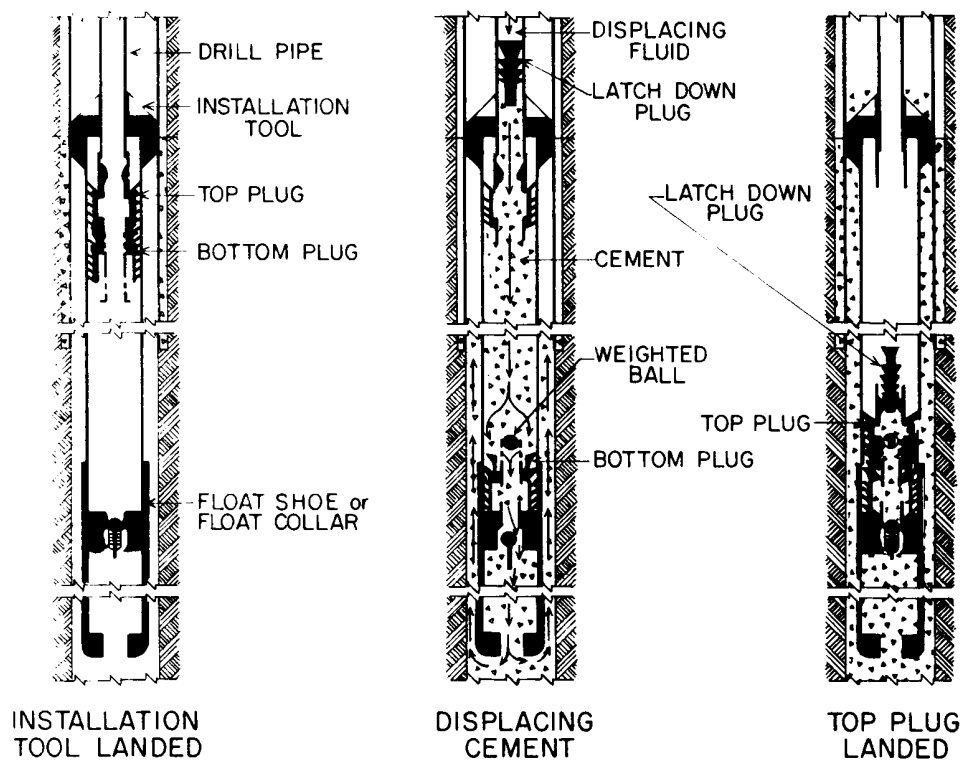


Fig. 3.10—Conventional placement techniques for cementing a liner.

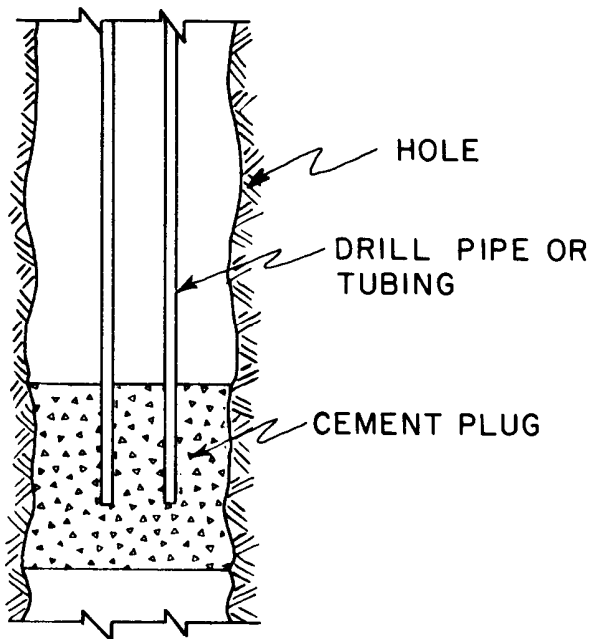
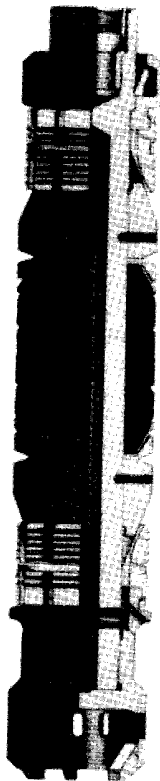


Fig. 3.11 — Placement technique used for setting cement plug.



Courtesy of Halliburton Services

Fig. 3.12 — Bridge plug.

plug, the inner string can be withdrawn immediately after displacing the cement.

3.5.4 Annular Cementing Through Tubing. This technique consists of pumping cement through tubing run in the annulus between two casing strings or between the casing and the open hole. It usually is used to bring the top of the previously placed cement to the surface or to repair casing.

3.5.5 Multiple-String Cementing. Multiple-string cementing is a multiple completion method that involves cementing several strings of tubing in the hole without the use of an outer casing string. This type of completion is an alternative to the more conventional multiple-completion method in which the tubing strings are set using packers inside a larger-diameter casing.

3.5.6 Reverse-Circulation Cementing. Reverse-circulation cementing consists of pumping the slurry down the annulus and displacing the mud back through the casing. This method has been used in some instances when extremely low-strength formations were present near the bottom of the hole. A special float collar or shoe as well as a special wellhead assembly is required to use this method. Since wiper plugs cannot be used, it is difficult to detect the end of the cement displacement. The cement usually is overdisplaced at least 300 ft into the bottom of the casing to enhance the probability of a good cement job at the shoe.

3.5.7 Delayed-Setting Cementing. Delayed-setting cementing is used to obtain a more uniform mud displacement from the annulus than is possible using the conventional cement placement technique. This method consists of placing a retarded cement slurry having good filtration properties in the wellbore before running the casing. Cement placement is accomplished down the drillpipe and up the annulus. The drillpipe then is removed from the well and casing is lowered into the unset cement slurry. This method also can be modified for multiple-string cementing with cement displacement being made through the lower string. The delayed-setting cementing method requires a drilling engineer with a strong nervous system.

3.5.8 Cementing Liners. The conventional method of liner cementing is illustrated in Fig. 3.10. The liner is attached to drillpipe using a special liner-setting tool. The liner is lowered to a position with several hundred feet of overlap between the top of the liner and the bottom of the casing. The liner-setting tool then is actuated so that the liner is attached mechanically to and supported by the casing without hydraulically sealing the passage between the liner and casing. The cement is pumped down the drillpipe, separated from the displacing fluid by a latch-down plug. This latch-down plug actuates a special wiper plug in the liner-setting tool after the top of the cement column reaches the liner. When the wiper plug reaches the float collar, a pressure in-

crease at the surface signifies the end of the cement displacement. The drillstring then must be released from the liner-setting tool and withdrawn before the cement hardens.

Usually, a volume of cement sufficient to extend past the top of the liner is displaced. When the drillstring is withdrawn, this cement collects at the top of the liner but generally does not fall to bottom. This cement can be washed out using the drillpipe or drilled out after the cement sets. When cement is not displaced to the top of the liner, cement must be forced into this area using a squeeze-cementing method, which is discussed in Sec. 3.5.10.

A large variety of liner-setting tools is available. Most are set with either a mechanical or hydraulic device. The hydraulically set devices are actuated by drillpipe rotation or by dropping a ball or plug and then set by applying pump pressure. The mechanically set devices are actuated by drillpipe rotation and set by lowering the drillpipe. A liner-setting tool must be selected on the basis of the liner weight, annular dimensions, cement displacement rate, and liner-cementing procedure. There is a trend toward not setting the liner until after cement displacement. This allows the liner to be moved while cementing to improve mud removal in the annulus. This practice imposes some limitations on the liner-setting tool selected.

After drilling the well, it may be desirable to extend the liner back to the surface if a commercial hydrocarbon deposit is found and the well is to be completed. This type of casing is called a *tieback liner*. If casing is used to extend the liner up the hole, but not all the way to the surface, it is called a *stub liner*. Stub liners are used primarily to repair the top of a leaking liner.

3.5.9 Plug Cementing. Cement plugs can be set in open hole or in casing. Plugs are set to prevent fluid communication between an abandoned lower portion of the well and the upper part of the well. Plugs also are set to provide a seat for directional drilling tools used to sidetrack the well.

Cement plugs are placed using drillpipe or tubing as shown in Fig. 3.11. When a plug is placed off bottom in an open hole, a caliper log can be used to locate an in-gauge portion of the hole. Centralizers and scratchers can be placed on the bottom section of the pipe that will be opposite the section of hole to be plugged. Cement is pumped down the drillpipe or tubing and into the hole. The cement slurry has a natural tendency to form a bridge below the tubing, causing the slurry to move up the annular space opposite the drillpipe or tubing. When cementing in casing, a bridge plug (Fig. 3.12) sometimes is placed below the cement plug to assist in forming a good hydraulic seal.

When the drillpipe or tubing is open-ended, cement displacement is continued until the fluid columns are *balanced* — i.e., have the same height of slurry inside the pipe and annulus. As shown in Fig. 3.13, the pump pressure during the displacement provides a good indication of when the fluid columns are balanced. The drillstring or tubing then is pulled slowly from the cement slurry.

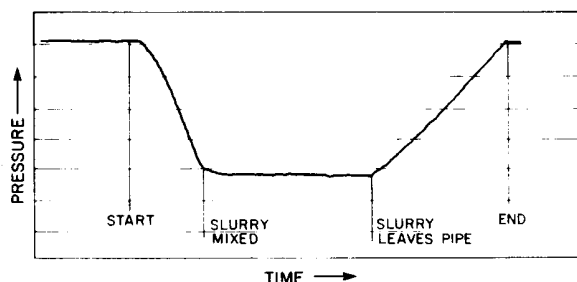


Fig. 3.13 – Idealized pressure/time chart for balanced pressure plug.

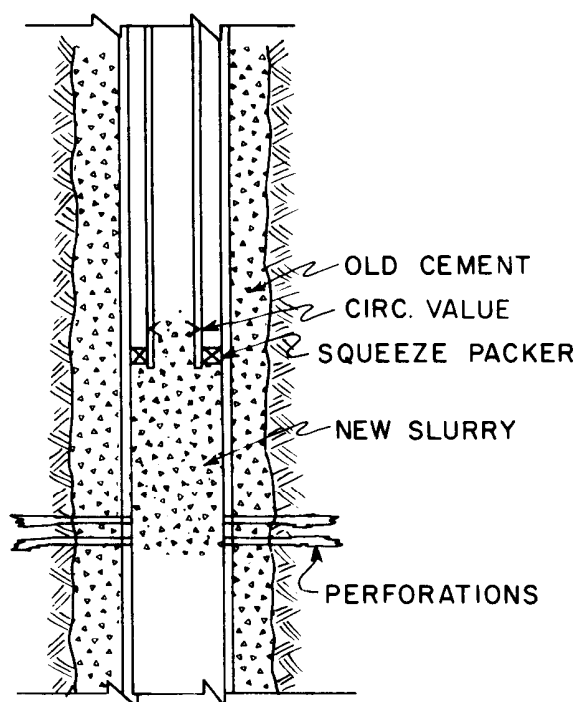


Fig. 3.14—Squeeze-cementing technique.

An improved plug placement technique has been described by Doherty⁸ and Goins⁹ as an alternative to the balanced-column method. Mud contamination is minimized by placing a slug of water in front of and behind the cement slurry. The cement is displaced almost completely from the drillpipe or tubing. To prevent backflow while the pipe is being pulled, a special backpressure valve or plug catcher can be used at the bottom of the pipe string. Cement plugs also can be set using a wireline device called a *dump bailer*. After setting a bridge plug at the desired depth, the dump bailer is filled with cement and lowered into the well on a wireline. The dump bailer is designed to empty the slurry into the hole when the bailer reaches bottom. After the first batch of cement takes an initial set, a second batch can be placed. The dump bailer seldom is used during drilling because of the time required to set a long plug.

3.5.10 Squeeze Cementing. Squeeze cementing consists of forcing a cement slurry into an area of the

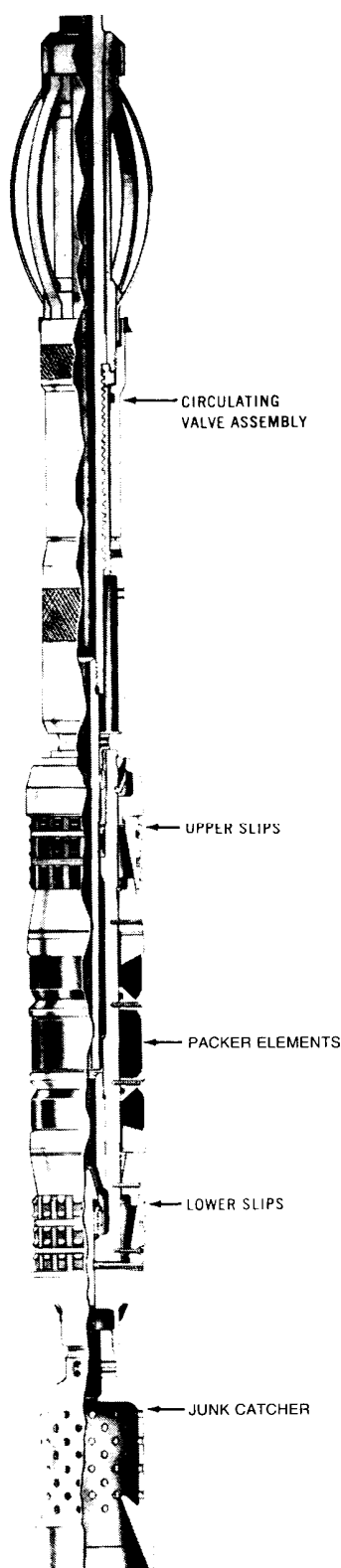


Fig. 3.15 – Squeeze packer assembly.⁵

well or formation by means of an applied hydraulic pressure. The purpose of squeeze cementing is to form a hydraulic seal between the wellbore and the zone squeezed. If the slurry is placed using sufficient pressure to fracture the formation, the process is called a *high-pressure squeeze*. The pressure required to fracture the formation and allow rapid slurry placement is called the *breakdown pressure*. In the high-pressure squeeze, whole cement slurry is forced into the fractured formation. If the slurry is placed using less than the breakdown pressure, the process is called a *low-pressure squeeze*. The low-pressure squeeze causes a cement cake to plate out against the formation as filtrate is lost to the formation. Common applications of squeeze cementing are (1) plugging abandoned casing perforations, (2) plugging severe lost-circulation zones, and (3) repairing annular leaks in previously cemented casing, caused by failure of either the casing or previously placed cement.

The conventional method for squeeze cementing is shown in Fig. 3.14. A squeeze packer and a circulating valve (Fig. 3.15) are placed above the perforations and lowered into the well on drillpipe or tubing. A slug of water or preflush is placed opposite the zone to be squeezed, and then the squeeze packer is set just above the zone of interest. The circulating valve is opened above the packer, and the cement slurry is displaced down the drillpipe. A slug of water is used before and after the cement slurry to prevent mud contamination of the slurry and mud plugging of the zone of interest. The circulating valve usually is closed after most of the leading water slug has been pumped into the annulus. The cement then is pumped into the zone of interest until the final desired squeeze pressure is obtained. With the low-pressure squeeze, it is common practice to stop pumping periodically, or *hesitate*, during the squeeze process. This assists buildup of filter cake nodes against the formation or perforations. The circulating valve then is opened, and the excess cement is pumped up the drillpipe to the surface. This process is called *reversing out*. If the desired squeeze pressure is not obtained, the operation is repeated after waiting for the first batch of cement to take an initial set.

In addition to the conventional squeeze-cementing method, several modified techniques have been developed. In certain instances, it may be necessary to isolate the section below the perforation by placing a bridge plug below the perforations. In other cases, neither a squeeze packer nor a bridge plug is used and pressure is held against the well by closing the blowout preventers. This method is called the *bradenhead squeeze*. A *block squeeze* consists of isolating a production zone by perforating above and below the producing interval and squeezing through both perforated intervals in separate steps.

3.5.11 Cement Volume Requirements. In addition to selecting the cement composition and placement technique, the drilling engineer must determine the volume of cement slurry needed for the job. The volume required usually is based on past experience

and regulatory requirements in the area. As little as 300 ft of fillup has been used behind relatively deep casing strings. However, in some cases, the entire annulus is cemented. It usually is necessary to include considerably more slurry than indicated by the theoretical hole size because of hole enlargement while drilling. Thus, an *excess factor*, based on prior experience in the area, usually is applied to the theoretical cement volume. When hole size or *caliper logs* are available, a more accurate slurry volume can be determined. Example 3.6 illustrates one method of estimating slurry volume requirements for conventional casing cementing operation.

Example 3.6. Casing having an OD of 13.375 in. and an ID of 12.415 in. is to be cemented at a depth of 2,500 ft. A 40-ft shoe joint will be used between the float collar and the guide shoe. It is desired to place a 500-ft column of high-strength slurry at the bottom of the casing. The high-strength slurry is composed of Class A cement mixed using a 2% calcium chloride flake (by weight of cement) and a water/cement ratio of 5.2 gal/sack. The upper 2,000 ft of the annulus is to be filled with a low-density slurry of Class A cement mixed with 16% bentonite and 5% sodium chloride (by weight of cement) and a water/cement ratio of 13 gal/sack. Compute the slurry volume requirements if the excess factor in the annulus is 1.75. The bit size used to drill the hole is 17 in.

Solution. The specific gravity of Class A cement is 3.14, and the specific gravity of bentonite is 2.65 (Table 3.8). The specific gravity of the brines must be determined using Tables 2.3 and 2.4 in Chap. 2. The NaCl brine is made by adding 4.7 lbm [0.05 (94)] of NaCl to 108.4 lbm [13 (8.34)] of water. Interpolating in Table 2.3 for a weight fraction of 0.0415 [or $4.7/(108.4 + 4.7)$] gives a NaCl brine specific gravity of 1.0279. The CaCl_2 brine is made by adding 1.88 lbm [0.02 (94)] of CaCl_2 to 43.4 lbm [5.2 (8.34)] of water. Interpolating in Table 2.4 for a weight fraction of 0.0415 [or $1.88/(43.4 + 1.88)$] gives a CaCl_2 brine specific gravity of 1.0329.

The volume of each component in the low-density lead slurry per sack of cement is given by:

Component	Volume (cu ft)	
	94	
Cement	$\frac{94}{3.14 (62.4)}$	= 0.4797
Bentonite	$\frac{0.16 (94)}{2.65 (62.4)}$	= 0.0910
Salt water	$\frac{108.4 + 4.7}{1.0279 (62.4)}$	= 1.7633
Yield		= 2.334 cu ft/sack

The annular capacity for the 17-in. hole and 13.375-in. casing is given by

$$A_a = \frac{\pi}{4} (17^2 - 13.375^2) \frac{\text{sq ft}}{144 \text{ sq in.}}$$

$$= 0.6006 \text{ sq ft.}$$

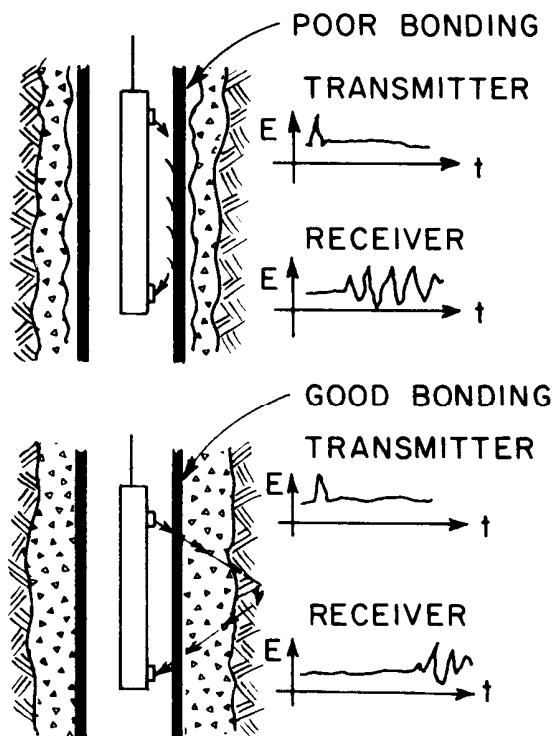


Fig. 3.16 – Acoustic energy travel in cased wells.

Using a column length of 2,000 ft and an excess factor of 1.75, the slurry volume required is

$$0.6006 (2,000) (1.75) = 2,102 \text{ cu ft.}$$

This slurry volume will require mixing

$$\frac{2,102 \text{ cu ft}}{2.334 \text{ cu ft/sack}} = 901 \text{ sacks.}$$

The volume of each component in the high-strength tail slurry is given by:

Component	Volume (cu ft)	
	94	
Cement	$\frac{94}{3.14 (62.4)}$	= 0.4797
CaCl_2 brine	$\frac{(43.4 + 1.88)}{1.0329 (62.4)}$	= 0.7025
Yield		= 1.182 cu ft/sack

The cement required for a 500-ft annular column and a 40-ft column in the shoe joint is given by

$$0.6006 (500) (1.75) + \frac{\pi}{4} (12.415)^2 \frac{40}{144}$$

$$= 559.2 \text{ cu ft.}$$

This slurry volume will require mixing

$$\frac{559.2 \text{ cu ft}}{1.182 \text{ cu ft/sack}} = 473 \text{ sacks of cement.}$$

Total slurry volume then will be

$$2,102 \text{ cu ft} + 559.2 \text{ cu ft} = 2,661.2 \text{ cu ft,}$$

and the total sacks of cement required for the job will be

$$901 + 473 = 1,374 \text{ sacks.}$$

3.5.12 Cementing Time Requirements. As discussed previously, the time required to place the cement slurry is one of the more important variables in the engineering design of the slurry properties. The relationships between well depth and cementing time used by the API in the specifications for the various cement classes (see Fig. 3.5) represent average well conditions and may not be applicable in all cases. A more accurate estimation of cementing time can be made based on the actual slurry volume and pumping rates to be used. Also, it is always prudent to allow some extra cementing time for unforeseen operational problems. Example 3.7 illustrates one method of estimating cementing time requirements for a conventional casing cementing operation.

Example 3.7. Estimate the cementing time for the cementing operation described in Example 3.6 if one cementing truck having a mixing capacity of approximately 20 cu ft/min is used. The rig pump will be operated at 60 strokes/min and has a pump factor of 0.9674 cu ft/stroke.

Solution. The volume of cement to be mixed is 901 sacks for the lead slurry and 473 sacks for the tail slurry. Each sack of cement has an approximate bulk volume of 1 cu ft. However, the lead slurry will be blended with 16% bentonite and 5% salt, and the tail slurry will be blended with 2% calcium chloride. The bulk weight of bentonite, salt, and calcium chloride (flake) are 60, 71, and 56.4 lbm/cu ft, respectively (Table 3.8). Thus, the additional volume of solids to be mixed is given by

$$\frac{901(94)(0.16)}{60} + \frac{901(94)(0.05)}{71} + \frac{473(94)(0.02)}{56.4} = 301 \text{ cu ft.}$$

The total mixing time is given by

$$\frac{1,374 + 301}{20} = 83.8 \text{ minutes.}$$

The mixing time accounts for the time period before placing the top plug at the top of the casing. The time required to displace the top plug from the surface to the float collar is given by

$$\frac{\pi}{4} (12.415)^2 \frac{(2,500 - 40)}{144} \times \frac{1}{60(0.9674)} = 35.6 \text{ minutes.}$$

Thus, the total cementing time is $83.8 + 35.6 = 119.4$ minutes or 2 hours.

3.5.13 Cementing Evaluation. After cementing operations are complete and the cement left in the

wellbore, along with portions of the subsurface cementing equipment, have been drilled out, the cement job usually is evaluated to ensure that the cementing objectives have been accomplished. It always is a good practice to pressure-test cemented casing to the maximum pressure anticipated in subsequent drilling operations. The top of the cement can be located by making a temperature survey of the well from 6 to 10 hours after completing the cement displacement. When cement is present behind the pipe, heat liberated due to the exothermic hydration reaction will cause an increase in temperature. In addition, acoustic logging tools are available for evaluating the bond between the cement and the pipe. When the cement is not bonded acoustically to both the pipe and the formation, a strong early sound reflection will be received by the acoustic logging device, indicating sound travel primarily through the casing (Fig. 3.16).

Exercises

3.1 List four steps in the manufacture of portland cement. What is the approximate weight and bulk volume of portland cement sold in 1 bbl and in one sack?

3.2 Name four crystalline phases in portland cement that hydrate when mixed with water. Show the chemical equation for the hydration of each phase. Identify the reaction product that is the main cementing ingredient in portland cement. Identify the crystalline phase that forms reaction products readily attacked by water-containing sulfates.

3.3 The oxide analysis of a cement is as follows.

Oxide	Weight Percent
Lime (CaO or C)	66.5
Silica (SiO ₂ or S)	21.1
Alumina (Al ₂ O ₃ or A)	4.8
Ferric oxide (Fe ₂ O ₃ or F)	2.6
Magnesia (MgO)	1.2
Sulfur trioxide (SO ₃)	2.7
Ignition loss	0.9

Calculate the theoretical weight fraction of C₃S, C₂S, C₃A, and C₄AF present in the cement. *Answer:* 0.666, 0.103, 0.083, 0.079.

3.4 List the equipment needed to perform the standard API tests for drilling cements.

3.5 Define the following: minimum water content, normal water content, free water content, and maximum water content.

3.6 The torque required to hold the paddle assembly stationary in a cement consistometer rotating at 150 rpm is 800 g-cm. What is the slurry consistency? *Answer:* 36 Bc.

3.7 A Class H cement core sample having a length of 2.54 cm and a diameter of 2.865 cm allows a water flow rate of 0.05 mL/s when placed under a pressure differential of 20 psi. Compute the permeability of the cement. *Answer:* 14.5 md.

3.8 The cement tensile strength required to support the weight of a string of casing is estimated to be 8 psi. If the cement is known to have a compressive strength of 200 psi, do you think the casing could be

supported by the cement? Why? *Answer:* Yes, tensile strength ≈ 17 psi.

3.9 List the eight standard classes and three standard types of API cement. Compare these classes and types with the ASTM cement types used in the construction industry.

3.10 How is the composition of a high-sulfate-resistant cement different from a standard portland cement?

3.11 List the normal (API) water content of each class of cement used in slurry preparation. *Answer:* 5.19 gal/sack for Classes A and B, 6.32 gal/sack for Class C, 4.29 gal/sack for Classes D through H, 4.97 gal/sack for Class G.

3.12 Compute the yield and density of each class of cement when mixed with the normal amount of water as defined by API. *Answer:* 1.17 cf/sack and 15.6 lbm/gal for Classes A and B.

3.13 It is desired to reduce the density of a Class A cement to 12.8 lbm/gal by adding bentonite. Using the water requirements for Class A cement and bentonite given in Table 3.9, compute the weight of bentonite that should be blended with each sack of cement. Compute the yield of the slurry. What is the "percent mix" of the slurry? *Answer:* 9.3 lbm/sack, 2.09 cf/sack, 102.9%.

3.14 Repeat Exercise 3.13 using diatomaceous earth instead of bentonite as the low-specific-gravity solid and a water requirement of 3.3 gal/10% diatomaceous earth. *Answer:* 19 lbm/sack, 2.21 cf/sack, 105%.

3.15 Identify the following cement additives: gilsonite, expanded perlite, and pozzolan.

3.16 It is desired to increase the density of a Class H cement to 17.5 lbm/gal using barite. Compute the weight of barite that should be blended with each sack of cement. Use the water requirements for Class H cement (maximum strength) and barite given in Table 3.9. Compute the yield of the slurry. What is the "percent mix" of the slurry? *Answer:* 29.4 lbm/sack, 1.26 cf/sack, 44.3%.

3.17 Repeat Problem 3.16 using sand instead of barite. *Answer:* 40.6 lbm/sack, 1.30 cf/sack, 38.2%.

3.18 List two common cement accelerators for Class A cement.

3.19 List two common cement retarders for Class H cement.

3.20 List the three types of lost-circulation additives used in cement, and give one example of each type.

3.21 List four common filtration-control additives.

3.22 Describe in your own words the conventional cement placement techniques used for cementing a casing string, cementing a liner string, setting a cement plug, and squeeze cementing.

3.23 What is the purpose of a shoe joint? Why is a shoe joint especially important in a deep cementing job or when no bottom wiper plug is used?

3.24 Casing having an OD of 9.625 in. and an ID of 8.535 in. is to be cemented at a depth of 13,300 ft in a 12.25-in. borehole. A 40-ft shoe joint will be used between the float collar and the guide shoe. It is desired to place 2,500 ft of cement in the annulus. Each sack of Class H cement will be mixed with 4.3

gal of water to which is added 18% salt (by weight of water). A small quantity of dispersant will be blended with the cement, but this additive has no significant effect on the slurry yield or density. Compute the density of the slurry, the yield of the slurry, the number of sacks of cement required, and the cementing time. Assume that cement can be mixed at a rate of 20 sacks/min and displaced at a rate of 9 bbl/min and use an excess factor of 1.5. The salt will be added to the water phase and thus does not blend with the dry cement. *Answer:* 16.7 lbm/gal; 1.09 cf/sack; 1,092 sacks; 159 minutes.

3.25 A 7.0-in. liner having an ID of 6.276 in. is to be cemented at a depth of 15,300 ft in an 8.5-in. hole. Casing is set at 13,300 ft as described in Problem 3.24, and a 300-ft overlap between the casing and liner is desired. A 40-ft shoe joint will be used between the float collar and the guide shoe. It is desired to use 1,000 ft of preflush (in annulus) and then fill the total annular space opposite the liner with cement. Class H cement containing 35% silica flour, and 1.2% HALAD-22A will be mixed with 5.8 gal of water containing 18% salt (by weight of water). Compute the slurry density, the slurry yield, the volume of cement slurry required if the caliper log shows an average washout of 2.0-in. increase in hole diameter, and the number of sacks of Class H cement needed. Neglect the effect of the HALAD-22A on the density and volume of the slurry. *Answer:* 16.3 lbm/gal, 1.51 cf/sack, 474 sacks.

Nomenclature

- A = area
- B_c = slurry consistency
- k = permeability
- L = length
- p = pressure
- q = flow rate
- T = torque
- μ = viscosity
- ρ = slurry density

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SI Metric Conversion Factors

bbl	×	1.589 873	E-01	=	m ³
cp	×	1.0*	E-03	=	Pa·s
cu ft	×	2.831 685	E-02	=	m ³
cu ft/lbm	×	6.242 796	E-02	=	m ³ /kg
ft	×	3.048*	E-01	=	m
°F	(°F-32) 1.8			=	°C
gal	×	3.785 412	E-03	=	m ³
gal/cu ft	×	1.336 806	E+02	=	dm ³ /m ³

gal/lbm	×	8.345 404	E+00	=	dm ³ /kg
in.	×	2.54*	E+00	=	cm
lbm	×	4.535 924	E-01	=	kg
lbm/cu ft	×	1.601 846	E+01	=	kg/m ³
lbm/gal	×	1.198 264	E+02	=	kg/m ³
psi	×	6.894 757	E+00	=	kPa
sq ft	×	9.290 304	E-02	=	m ²
sq in.	×	6.451 6*	E+00	=	cm ²

*Conversion factor is exact.